missionML.R

imrah

2026-10-31

```
missionML <- read.csv("C:/Users/imrah/Downloads/missionML.csv", stringsAsFactors=TRUE)
summary(missionML)
```

```
##       SL.              AGE        GENDER    MARITAL.STATUS
##  Min.   :  1.00   Min.   : 0.03   F: 82   MARRIED  :108
##  1st Qu.: 62.75   1st Qu.: 6.00   M:166   UNMARRIED:140
##  Median :124.50   Median :15.50
##  Mean   :124.50   Mean   :28.88
##  3rd Qu.:186.25   3rd Qu.:55.00
##  Max.   :248.00   Max.   :88.00
##
##    KEY.COMPLAINTS..CODE  BODY.WEIGHT     BODY.HEIGHT         BMI
##  other- heart:55        Min.   : 2.00   Min.   : 19.0   Min.   :0.04489
##               :36        1st Qu.:15.00   1st Qu.:105.0   1st Qu.:0.15142
##  CAD-DVD      :27        Median :41.00   Median :147.5   Median :0.26473
##  RHD          :26        Mean   :37.54   Mean   :130.2   Mean   :0.26723
##  CAD-TVD      :24        3rd Qu.:58.25   3rd Qu.:160.0   3rd Qu.:0.37164
##  ACHD         :19        Max.   :85.00   Max.   :185.0   Max.   :0.76842
##  (Other)      :61
##     HR.PULSE         BP..HIGH       BP.LOW             RR
##  Min.   : 41.00   Min.   : 70   Min.   : 39.00   Min.   :12.00
##  1st Qu.: 78.00   1st Qu.:100   1st Qu.: 60.00   1st Qu.:22.00
##  Median : 90.00   Median :110   Median : 70.00   Median :24.00
##  Mean   : 92.23   Mean   :115   Mean   : 71.88   Mean   :23.54
##  3rd Qu.:104.00   3rd Qu.:130   3rd Qu.: 80.00   3rd Qu.:24.00
##  Max.   :155.00   Max.   :215   Max.   :140.00   Max.   :42.00
##                   NA's   :23    NA's   :23
##  PAST.MEDICAL.HISTORY.CODE       HB              UREA          CREATININE
##               :173         Min.   : 5.00   Min.   :  2.00   Min.   :0.100
##  hypertension1: 20         1st Qu.:11.20   1st Qu.: 18.00   1st Qu.:0.300
##  other        : 15         Median :12.30   Median : 22.00   Median :0.700
##  hypertension2: 13         Mean   :12.88   Mean   : 26.58   Mean   :0.747
##  Diabetes1    : 10         3rd Qu.:14.00   3rd Qu.: 30.00   3rd Qu.:1.000
##  Diabetes2    :  9         Max.   :25.70   Max.   :143.00   Max.   :5.200
##  (Other)      :  8                         NA's   :13       NA's   :33
##     MODE.OF.ARRIVAL STATE.AT.THE.TIME.OF.ARRIVAL   TYPE.OF.ADMSN
##  AMBULANCE  : 30    ALERT   :247                 ELECTIVE :216
##  TRANSFERRED:  4    CONFUSED:  1                 EMERGENCY: 32
##  WALKED IN  :214
##
##
##
##
##  TOTAL.COST.TO.HOSPITAL TOTAL.LENGTH.OF.STAY LENGTH.OF.STAY...ICU
##  Min.   : 46093         Min.   : 3.00        Min.   : 0.000
##  1st Qu.:131653         1st Qu.: 8.00        1st Qu.: 1.000
##  Median :162661         Median :10.00        Median : 2.000
##  Mean   :198723         Mean   :11.61        Mean   : 3.476
##  3rd Qu.:220614         3rd Qu.:13.00        3rd Qu.: 4.000
##  Max.   :887350         Max.   :41.00        Max.   :30.000
##
##  LENGTH.OF.STAY..WARD IMPLANT.USED..Y.N. COST.OF.IMPLANT
##  Min.   : 0.000       N:199              Min.   :     0
##  1st Qu.: 6.000       Y: 49              1st Qu.:     0
##  Median : 7.000                          Median :     0
##  Mean   : 8.153                          Mean   :  8544
##  3rd Qu.:10.000                          3rd Qu.:     0
```

```
##  Max.   :22.000                         Max.   :196848
##
```

```
#Missing Values
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✓ dplyr     1.1.4     ✓ readr     2.1.6
## ✓ forcats   1.0.1     ✓ stringr   1.6.0
## ✓ ggplot2   4.0.1     ✓ tibble    3.3.0
## ✓ lubridate 1.9.4     ✓ tidyr     1.3.1
## ✓ purrr     1.2.0
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to be
come errors
```

```
missionML <- missionML %>%
  mutate(across(where(is.numeric),
                ~ ifelse(is.na(.), mean(., na.rm = TRUE), .)))
#Label Encoding
missionML$TYPE.OF.ADMSN <- ifelse(missionML$TYPE.OF.ADMSN=="EMERGENCY",1,0)
missionML$IMPLANT.USED..Y.N. <- ifelse(missionML$IMPLANT.USED..Y.N.=="Y",1,0)

#Dummy variables
library(fastDummies)
missionML = dummy_cols(
  missionML,
  select_columns = c(
    "KEY.COMPLAINTS..CODE",
    "PAST.MEDICAL.HISTORY.CODE",
    "MODE.OF.ARRIVAL"))

#correlation
cr = cor(missionML[sapply(missionML,is.numeric)])
cr
```

```
##                                              SL.         AGE BODY.WEIGHT
## SL.                                       1.000000000 -0.32423547 -0.24409918
## AGE                                      -0.324235470  1.00000000  0.84844828
## BODY.WEIGHT                              -0.244099185  0.84844828  1.00000000
## BODY.HEIGHT                              -0.216327198  0.71749067  0.85021339
## BMI                                      -0.188977419  0.73127999  0.86153236
## HR.PULSE                                  0.058443305 -0.48349857 -0.56573983
## BP..HIGH                                 -0.123438006  0.57885819  0.57297297
## BP.LOW                                   -0.113011906  0.44565999  0.47499276
## RR                                       -0.121831164 -0.28437685 -0.31729045
## HB                                        0.103041281 -0.12920949 -0.08630186
## UREA                                     -0.170278064  0.22696788  0.16798070
## CREATININE                               -0.404548625  0.57156016  0.54944385
## TYPE.OF.ADMSN                            -0.248489923  0.46668129  0.35228042
## TOTAL.COST.TO.HOSPITAL                   -0.498936977  0.42069454  0.34835849
## TOTAL.LENGTH.OF.STAY                     -0.407325319  0.21510574  0.05352518
## LENGTH.OF.STAY...ICU                     -0.448393194  0.34517759  0.22840200
## LENGTH.OF.STAY..WARD                     -0.120289533 -0.05182204 -0.16001717
## IMPLANT.USED..Y.N.                       -0.034727170  0.20738253  0.26406963
## COST.OF.IMPLANT                          -0.099933285  0.18217005  0.27759450
## KEY.COMPLAINTS..CODE_                    -0.219849757  0.07587382  0.13576652
## KEY.COMPLAINTS..CODE_ACHD                 0.108951524 -0.12906520 -0.15714295
## KEY.COMPLAINTS..CODE_CAD-DVD             -0.221423261  0.42234153  0.33914083
## KEY.COMPLAINTS..CODE_CAD-SVD             -0.076198302  0.11902738  0.08582236
## KEY.COMPLAINTS..CODE_CAD-TVD             -0.111828099  0.38029588  0.41616919
## KEY.COMPLAINTS..CODE_CAD-VSD              0.026218993 -0.06370910 -0.06765622
## KEY.COMPLAINTS..CODE_OS-ASD               0.221508379 -0.12119929 -0.06893765
## KEY.COMPLAINTS..CODE_other- heart         0.097140127 -0.21071744 -0.27445120
## KEY.COMPLAINTS..CODE_other- respiratory -0.097227411 -0.19278765 -0.22434071
## KEY.COMPLAINTS..CODE_other-general        0.030662890  0.11107352  0.07297088
## KEY.COMPLAINTS..CODE_other-nervous       -0.006955568 -0.11793222 -0.14323681
## KEY.COMPLAINTS..CODE_other-tertalogy      0.143497311 -0.24708274 -0.24223177
## KEY.COMPLAINTS..CODE_PM-VSD               0.161291634 -0.14961730 -0.18676009
## KEY.COMPLAINTS..CODE_RHD                  0.073358798  0.04029345  0.15803468
## PAST.MEDICAL.HISTORY.CODE_                0.207547481 -0.49549922 -0.42259172
## PAST.MEDICAL.HISTORY.CODE_Diabetes1      -0.008303324  0.23252318  0.23418838
## PAST.MEDICAL.HISTORY.CODE_Diabetes2      -0.239887737  0.21097775  0.24497158
## PAST.MEDICAL.HISTORY.CODE_hypertension1  -0.164447464  0.37836864  0.28318707
## PAST.MEDICAL.HISTORY.CODE_Hypertension1   0.013653521  0.13751297  0.10452555
## PAST.MEDICAL.HISTORY.CODE_hypertension2  -0.037275964  0.15018818  0.16702399
## PAST.MEDICAL.HISTORY.CODE_hypertension3  -0.169710123  0.17936750  0.13591050
## PAST.MEDICAL.HISTORY.CODE_other           0.111640223 -0.14410443 -0.17889264
## MODE.OF.ARRIVAL_AMBULANCE                -0.242505005  0.44903716  0.35362584
## MODE.OF.ARRIVAL_TRANSFERRED               0.001788453  0.03031701  0.02476728
## MODE.OF.ARRIVAL_WALKED IN                 0.229257823 -0.43682437 -0.34433473
##                                           BODY.HEIGHT         BMI     HR.PULSE
## SL.                                      -0.216327198 -0.18897742  0.058443305
## AGE                                       0.717490669  0.73127999 -0.483498571
## BODY.WEIGHT                               0.850213386  0.86153236 -0.565739830
## BODY.HEIGHT                               1.000000000  0.54582796 -0.535003825
## BMI                                       0.545827964  1.00000000 -0.510257969
## HR.PULSE                                 -0.535003825 -0.51025797  1.000000000
## BP..HIGH                                  0.452728835  0.51078197 -0.294296478
## BP.LOW                                    0.384749101  0.45474062 -0.202306597
## RR                                       -0.300502674 -0.29054752  0.398739336
```

```
## HB                                       -0.056944410 -0.11726296  0.032780068
## UREA                                      0.158557012  0.11367561 -0.023677432
## CREATININE                                0.493724422  0.43897029 -0.358653510
## TYPE.OF.ADMSN                             0.274428788  0.31336286 -0.149169406
## TOTAL.COST.TO.HOSPITAL                    0.293827587  0.26634523 -0.008536208
## TOTAL.LENGTH.OF.STAY                      0.002190531  0.03020662  0.123732721
## LENGTH.OF.STAY...ICU                      0.137964880  0.18146600  0.041948703
## LENGTH.OF.STAY..WARD                     -0.139424678 -0.14444874  0.136956581
## IMPLANT.USED..Y.N.                        0.310380388  0.20400093 -0.134038362
## COST.OF.IMPLANT                           0.277072494  0.23320032 -0.073422918
## KEY.COMPLAINTS..CODE_                     0.168373850  0.09534623 -0.137959872
## KEY.COMPLAINTS..CODE_ACHD                -0.110608793 -0.17192102  0.117161056
## KEY.COMPLAINTS..CODE_CAD-DVD              0.253671342  0.30258643 -0.134903084
## KEY.COMPLAINTS..CODE_CAD-SVD              0.084830498  0.06317185 -0.096671470
## KEY.COMPLAINTS..CODE_CAD-TVD              0.293301004  0.35100128 -0.248655543
## KEY.COMPLAINTS..CODE_CAD-VSD             -0.060584620 -0.06617882  0.036951180
## KEY.COMPLAINTS..CODE_OS-ASD              -0.032161371 -0.01293197 -0.036260717
## KEY.COMPLAINTS..CODE_other- heart        -0.274406076 -0.18777546  0.100259548
## KEY.COMPLAINTS..CODE_other- respiratory -0.201348361 -0.21848758  0.225799101
## KEY.COMPLAINTS..CODE_other-general        0.040330678  0.07354203  0.030672391
## KEY.COMPLAINTS..CODE_other-nervous       -0.166694564 -0.13824770  0.038783234
## KEY.COMPLAINTS..CODE_other-tertalogy     -0.182083862 -0.22464424  0.167828751
## KEY.COMPLAINTS..CODE_PM-VSD              -0.272778456 -0.09068730  0.158772692
## KEY.COMPLAINTS..CODE_RHD                  0.245200590  0.09305403 -0.070771974
## PAST.MEDICAL.HISTORY.CODE_               -0.267999581 -0.37750833  0.224909127
## PAST.MEDICAL.HISTORY.CODE_Diabetes1       0.148802613  0.21636722 -0.073119718
## PAST.MEDICAL.HISTORY.CODE_Diabetes2       0.189187665  0.19438366 -0.043690149
## PAST.MEDICAL.HISTORY.CODE_hypertension1   0.233944969  0.23392891 -0.236437380
## PAST.MEDICAL.HISTORY.CODE_Hypertension1   0.108825340  0.07310327 -0.084970873
## PAST.MEDICAL.HISTORY.CODE_hypertension2   0.101410783  0.15594248 -0.074080441
## PAST.MEDICAL.HISTORY.CODE_hypertension3   0.079913133  0.13985121 -0.147421949
## PAST.MEDICAL.HISTORY.CODE_other          -0.213896757 -0.13265357  0.126483437
## MODE.OF.ARRIVAL_AMBULANCE                 0.264552158  0.31810295 -0.118296657
## MODE.OF.ARRIVAL_TRANSFERRED               0.016468864  0.02240091 -0.004610588
## MODE.OF.ARRIVAL_WALKED IN                -0.256846926 -0.30978969  0.113842692
##                                               BP..HIGH       BP.LOW           RR
## SL.                                      -1.234380e-01 -0.113011906 -0.121831164
## AGE                                       5.788582e-01  0.445659989 -0.284376846
## BODY.WEIGHT                               5.729730e-01  0.474992761 -0.317290448
## BODY.HEIGHT                               4.527288e-01  0.384749101 -0.300502674
## BMI                                       5.107820e-01  0.454740623 -0.290547519
## HR.PULSE                                 -2.942965e-01 -0.202306597  0.398739336
## BP..HIGH                                  1.000000e+00  0.766871932 -0.127063146
## BP.LOW                                    7.668719e-01  1.000000000 -0.072351384
## RR                                       -1.270631e-01 -0.072351384  1.000000000
## HB                                       -5.647060e-02  0.039693432  0.006586784
## UREA                                      5.094836e-02  0.024199729  0.039256886
## CREATININE                                3.631237e-01  0.189255417 -0.218747191
## TYPE.OF.ADMSN                             2.158111e-01  0.180158914 -0.022877342
## TOTAL.COST.TO.HOSPITAL                    1.767461e-01  0.152892192  0.016180748
## TOTAL.LENGTH.OF.STAY                      3.954904e-02  0.015222909  0.175509536
## LENGTH.OF.STAY...ICU                      1.332358e-01  0.084369651  0.055048725
## LENGTH.OF.STAY..WARD                     -8.619939e-02 -0.067710833  0.199682465
## IMPLANT.USED..Y.N.                        5.860833e-02  0.085483678  0.011953010
## COST.OF.IMPLANT                           3.498894e-02  0.078585523 -0.021318370
## KEY.COMPLAINTS..CODE_                     6.799261e-02  0.027808338  0.028514878
```

```
## KEY.COMPLAINTS..CODE_ACHD               -8.237462e-02 -0.125925134  0.105737245
## KEY.COMPLAINTS..CODE_CAD-DVD             2.293175e-01  0.204021626 -0.214764591
## KEY.COMPLAINTS..CODE_CAD-SVD            -9.355195e-05 -0.042598253 -0.036234295
## KEY.COMPLAINTS..CODE_CAD-TVD             2.798428e-01  0.248330999 -0.060374717
## KEY.COMPLAINTS..CODE_CAD-VSD            -7.433533e-02 -0.095641672 -0.025569597
## KEY.COMPLAINTS..CODE_OS-ASD             -1.502998e-01 -0.071199049 -0.066658140
## KEY.COMPLAINTS..CODE_other- heart       -1.382460e-01 -0.114061841  0.051359051
## KEY.COMPLAINTS..CODE_other- respiratory -5.551125e-02 -0.075992832  0.136341341
## KEY.COMPLAINTS..CODE_other-general       4.449557e-02 -0.008200078  0.007630718
## KEY.COMPLAINTS..CODE_other-nervous      -6.896306e-02 -0.060197271  0.013270627
## KEY.COMPLAINTS..CODE_other-tertalogy    -9.671897e-02 -0.037785179  0.009221211
## KEY.COMPLAINTS..CODE_PM-VSD             -1.471423e-01 -0.103690236  0.005190193
## KEY.COMPLAINTS..CODE_RHD                 2.425410e-02  0.038878290  0.027305665
## PAST.MEDICAL.HISTORY.CODE_              -3.050926e-01 -0.164882840  0.168398133
## PAST.MEDICAL.HISTORY.CODE_Diabetes1      1.911950e-01  0.170769345 -0.173284794
## PAST.MEDICAL.HISTORY.CODE_Diabetes2      1.558363e-01  0.078699941 -0.027354961
## PAST.MEDICAL.HISTORY.CODE_hypertension1  1.932874e-01  0.155164485 -0.088112188
## PAST.MEDICAL.HISTORY.CODE_Hypertension1  7.738252e-02 -0.014260806 -0.005975662
## PAST.MEDICAL.HISTORY.CODE_hypertension2  1.222402e-01  0.019415571 -0.085076756
## PAST.MEDICAL.HISTORY.CODE_hypertension3  1.404945e-01  0.080077791 -0.132490697
## PAST.MEDICAL.HISTORY.CODE_other          -1.455788e-01 -0.121032701  0.101037084
## MODE.OF.ARRIVAL_AMBULANCE                2.218897e-01  0.179902378  0.018679901
## MODE.OF.ARRIVAL_TRANSFERRED              4.470185e-02 -0.038495045  0.032056865
## MODE.OF.ARRIVAL_WALKED IN               -2.267401e-01 -0.156462042 -0.029450792
##                                                   HB          UREA  CREATININE
## SL.                                       0.103041281 -1.702781e-01 -0.40454863
## AGE                                      -0.129209491  2.269679e-01  0.57156016
## BODY.WEIGHT                              -0.086301856  1.679807e-01  0.54944385
## BODY.HEIGHT                              -0.056944410  1.585570e-01  0.49372442
## BMI                                      -0.117262956  1.136756e-01  0.43897029
## HR.PULSE                                  0.032780068 -2.367743e-02 -0.35865351
## BP..HIGH                                 -0.056470595  5.094836e-02  0.36312372
## BP.LOW                                    0.039693432  2.419973e-02  0.18925542
## RR                                        0.006586784  3.925689e-02 -0.21874719
## HB                                        1.000000000 -6.800397e-02 -0.16065332
## UREA                                     -0.068003973  1.000000e+00  0.45489936
## CREATININE                               -0.160653325  4.548994e-01  1.00000000
## TYPE.OF.ADMSN                            -0.056224476  4.075483e-01  0.47304996
## TOTAL.COST.TO.HOSPITAL                   -0.059844819  2.720978e-01  0.36752827
## TOTAL.LENGTH.OF.STAY                      0.014979136  2.196512e-01  0.17073646
## LENGTH.OF.STAY...ICU                     -0.086748619  2.321988e-01  0.26736349
## LENGTH.OF.STAY..WARD                      0.115414190  7.967623e-02 -0.03062676
## IMPLANT.USED..Y.N.                       -0.091377854  2.138407e-01  0.20110663
## COST.OF.IMPLANT                          -0.090055980  2.375169e-01  0.20302920
## KEY.COMPLAINTS..CODE_                    -0.078148042 -1.114892e-01  0.15132872
## KEY.COMPLAINTS..CODE_ACHD                -0.096843866 -5.106049e-05 -0.14711378
## KEY.COMPLAINTS..CODE_CAD-DVD             -0.035178883  5.689146e-02  0.23568676
## KEY.COMPLAINTS..CODE_CAD-SVD              0.006484456  8.075619e-02  0.06245702
## KEY.COMPLAINTS..CODE_CAD-TVD             -0.005521639  8.604233e-02  0.22389548
## KEY.COMPLAINTS..CODE_CAD-VSD             -0.036886498  5.807126e-03 -0.05580421
## KEY.COMPLAINTS..CODE_OS-ASD              -0.045132177 -8.179222e-02 -0.13291428
## KEY.COMPLAINTS..CODE_other- heart         0.094681176 -5.395369e-02 -0.13334723
## KEY.COMPLAINTS..CODE_other- respiratory -0.030802950 -4.868265e-02 -0.06069906
## KEY.COMPLAINTS..CODE_other-general       -0.065910184  3.456451e-01  0.14395280
## KEY.COMPLAINTS..CODE_other-nervous       -0.080974620  3.858182e-02 -0.06469963
## KEY.COMPLAINTS..CODE_other-tertalogy      0.403951405 -6.831385e-02 -0.17234708
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## KEY.COMPLAINTS..CODE_PM-VSD             -0.052804635 -2.278126e-02 -0.12779816
## KEY.COMPLAINTS..CODE_RHD                -0.121627946  1.212914e-01  0.02617497
## PAST.MEDICAL.HISTORY.CODE_               0.084911135 -1.114228e-01 -0.34552348
## PAST.MEDICAL.HISTORY.CODE_Diabetes1     -0.011305123 -8.549542e-02  0.05941336
## PAST.MEDICAL.HISTORY.CODE_Diabetes2     -0.072453193  1.814314e-01  0.26979941
## PAST.MEDICAL.HISTORY.CODE_hypertension1  0.003881730  7.589844e-02  0.30667786
## PAST.MEDICAL.HISTORY.CODE_Hypertension1 -0.052131609  5.352105e-03  0.05493746
## PAST.MEDICAL.HISTORY.CODE_hypertension2 -0.028651946 -1.394790e-02  0.11467235
## PAST.MEDICAL.HISTORY.CODE_hypertension3 -0.008378623 -3.306082e-02  0.05432779
## PAST.MEDICAL.HISTORY.CODE_other         -0.046234426  8.628832e-02 -0.10968129
## MODE.OF.ARRIVAL_AMBULANCE               -0.054745307  4.135389e-01  0.47161321
## MODE.OF.ARRIVAL_TRANSFERRED             -0.082568437  1.992449e-02  0.12009216
## MODE.OF.ARRIVAL_WALKED IN                0.082143392 -3.993631e-01 -0.49110837
##                                         TYPE.OF.ADMSN TOTAL.COST.TO.HOSPITAL
## SL.                                       -0.24848992           -0.498936977
## AGE                                        0.46668129            0.420694544
## BODY.WEIGHT                                0.35228042            0.348358487
## BODY.HEIGHT                                0.27442879            0.293827587
## BMI                                        0.31336286            0.266345232
## HR.PULSE                                  -0.14916941           -0.008536208
## BP..HIGH                                   0.21581114            0.176746081
## BP.LOW                                     0.18015891            0.152892192
## RR                                        -0.02287734            0.016180748
## HB                                        -0.05622448           -0.059844819
## UREA                                       0.40754835            0.272097833
## CREATININE                                 0.47304996            0.367528267
## TYPE.OF.ADMSN                              1.00000000            0.353420153
## TOTAL.COST.TO.HOSPITAL                     0.35342015            1.000000000
## TOTAL.LENGTH.OF.STAY                       0.29588407            0.698708979
## LENGTH.OF.STAY...ICU                       0.39024979            0.821994878
## LENGTH.OF.STAY..WARD                       0.02598280            0.160676151
## IMPLANT.USED..Y.N.                         0.08088009            0.423276129
## COST.OF.IMPLANT                            0.07851083            0.483825185
## KEY.COMPLAINTS..CODE_                      0.01211607           -0.180963411
## KEY.COMPLAINTS..CODE_ACHD                 -0.06564567           -0.165320348
## KEY.COMPLAINTS..CODE_CAD-DVD               0.32886225            0.318524329
## KEY.COMPLAINTS..CODE_CAD-SVD              -0.03470528            0.051958055
## KEY.COMPLAINTS..CODE_CAD-TVD               0.15879757            0.213478889
## KEY.COMPLAINTS..CODE_CAD-VSD              -0.02449061           -0.030258262
## KEY.COMPLAINTS..CODE_OS-ASD               -0.09765979           -0.119422675
## KEY.COMPLAINTS..CODE_other- heart         -0.08966013           -0.041382843
## KEY.COMPLAINTS..CODE_other- respiratory   -0.09765979           -0.094288709
## KEY.COMPLAINTS..CODE_other-general         0.16531163           -0.074289145
## KEY.COMPLAINTS..CODE_other-nervous        -0.04259177           -0.036708394
## KEY.COMPLAINTS..CODE_other-tertalogy      -0.10767638           -0.070798836
## KEY.COMPLAINTS..CODE_PM-VSD               -0.06060606           -0.069586023
## KEY.COMPLAINTS..CODE_RHD                  -0.05319542            0.158573732
## PAST.MEDICAL.HISTORY.CODE_                -0.21794863           -0.171344291
## PAST.MEDICAL.HISTORY.CODE_Diabetes1        0.16568337            0.087828392
## PAST.MEDICAL.HISTORY.CODE_Diabetes2        0.18257890            0.309590791
## PAST.MEDICAL.HISTORY.CODE_hypertension1    0.15104681            0.184849101
## PAST.MEDICAL.HISTORY.CODE_Hypertension1   -0.04259177           -0.021327816
## PAST.MEDICAL.HISTORY.CODE_hypertension2   -0.03655963           -0.093824027
## PAST.MEDICAL.HISTORY.CODE_hypertension3    0.11594427            0.049971669
## PAST.MEDICAL.HISTORY.CODE_other           -0.04720223           -0.128318173
## MODE.OF.ARRIVAL_AMBULANCE                  0.96379411            0.353738721
```

```
## MODE.OF.ARRIVAL_TRANSFERRED               -0.04928142          -0.075391827
## MODE.OF.ARRIVAL_WALKED IN                 -0.89569939          -0.307758394
##                                      TOTAL.LENGTH.OF.STAY
## SL.                                          -0.407325319
## AGE                                           0.215105742
## BODY.WEIGHT                                   0.053525177
## BODY.HEIGHT                                   0.002190531
## BMI                                           0.030206621
## HR.PULSE                                      0.123732721
## BP..HIGH                                      0.039549037
## BP.LOW                                        0.015222909
## RR                                            0.175509536
## HB                                            0.014979136
## UREA                                          0.219651242
## CREATININE                                    0.170736463
## TYPE.OF.ADMSN                                 0.295884065
## TOTAL.COST.TO.HOSPITAL                        0.698708979
## TOTAL.LENGTH.OF.STAY                          1.000000000
## LENGTH.OF.STAY...ICU                          0.722155404
## LENGTH.OF.STAY..WARD                          0.698175009
## IMPLANT.USED..Y.N.                            0.135422579
## COST.OF.IMPLANT                               0.107599429
## KEY.COMPLAINTS..CODE_                        -0.161785915
## KEY.COMPLAINTS..CODE_ACHD                    -0.122947399
## KEY.COMPLAINTS..CODE_CAD-DVD                  0.277715276
## KEY.COMPLAINTS..CODE_CAD-SVD                  0.090256815
## KEY.COMPLAINTS..CODE_CAD-TVD                  0.074228777
## KEY.COMPLAINTS..CODE_CAD-VSD                  0.004576662
## KEY.COMPLAINTS..CODE_OS-ASD                  -0.063469757
## KEY.COMPLAINTS..CODE_other- heart             0.045611269
## KEY.COMPLAINTS..CODE_other- respiratory      -0.041468259
## KEY.COMPLAINTS..CODE_other-general           -0.042715514
## KEY.COMPLAINTS..CODE_other-nervous            0.014813143
## KEY.COMPLAINTS..CODE_other-tertalogy         -0.037635465
## KEY.COMPLAINTS..CODE_PM-VSD                  -0.086201206
## KEY.COMPLAINTS..CODE_RHD                      0.009940849
## PAST.MEDICAL.HISTORY.CODE_                   -0.088140847
## PAST.MEDICAL.HISTORY.CODE_Diabetes1           0.060449500
## PAST.MEDICAL.HISTORY.CODE_Diabetes2           0.138157221
## PAST.MEDICAL.HISTORY.CODE_hypertension1       0.103853116
## PAST.MEDICAL.HISTORY.CODE_Hypertension1      -0.019456069
## PAST.MEDICAL.HISTORY.CODE_hypertension2      -0.070489404
## PAST.MEDICAL.HISTORY.CODE_hypertension3       0.020979145
## PAST.MEDICAL.HISTORY.CODE_other              -0.044611330
## MODE.OF.ARRIVAL_AMBULANCE                     0.277129450
## MODE.OF.ARRIVAL_TRANSFERRED                  -0.032424843
## MODE.OF.ARRIVAL_WALKED IN                    -0.250863779
##                                      LENGTH.OF.STAY...ICU
## SL.                                          -0.44839319
## AGE                                           0.34517759
## BODY.WEIGHT                                   0.22840200
## BODY.HEIGHT                                   0.13796488
## BMI                                           0.18146600
## HR.PULSE                                      0.04194870
## BP..HIGH                                      0.13323580
## BP.LOW                                        0.08436965
```

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## RR                                                0.05504872
## HB                                               -0.08674862
## UREA                                              0.23219876
## CREATININE                                        0.26736349
## TYPE.OF.ADMSN                                     0.39024979
## TOTAL.COST.TO.HOSPITAL                            0.82199488
## TOTAL.LENGTH.OF.STAY                              0.72215540
## LENGTH.OF.STAY...ICU                              1.00000000
## LENGTH.OF.STAY..WARD                              0.01032756
## IMPLANT.USED..Y.N.                                0.14663485
## COST.OF.IMPLANT                                   0.15873838
## KEY.COMPLAINTS..CODE_                            -0.12241907
## KEY.COMPLAINTS..CODE_ACHD                        -0.13024753
## KEY.COMPLAINTS..CODE_CAD-DVD                      0.32030614
## KEY.COMPLAINTS..CODE_CAD-SVD                     -0.02287868
## KEY.COMPLAINTS..CODE_CAD-TVD                      0.21129798
## KEY.COMPLAINTS..CODE_CAD-VSD                     -0.04096264
## KEY.COMPLAINTS..CODE_OS-ASD                      -0.11056338
## KEY.COMPLAINTS..CODE_other- heart                 0.02985851
## KEY.COMPLAINTS..CODE_other- respiratory          -0.04458716
## KEY.COMPLAINTS..CODE_other-general               -0.02441747
## KEY.COMPLAINTS..CODE_other-nervous                0.01508305
## KEY.COMPLAINTS..CODE_other-tertalogy             -0.10735481
## KEY.COMPLAINTS..CODE_PM-VSD                      -0.06042506
## KEY.COMPLAINTS..CODE_RHD                         -0.05945394
## PAST.MEDICAL.HISTORY.CODE_                       -0.18790681
## PAST.MEDICAL.HISTORY.CODE_Diabetes1               0.11322058
## PAST.MEDICAL.HISTORY.CODE_Diabetes2               0.26753429
## PAST.MEDICAL.HISTORY.CODE_hypertension1           0.12123429
## PAST.MEDICAL.HISTORY.CODE_Hypertension1           0.02467431
## PAST.MEDICAL.HISTORY.CODE_hypertension2          -0.06673571
## PAST.MEDICAL.HISTORY.CODE_hypertension3           0.10161074
## PAST.MEDICAL.HISTORY.CODE_other                  -0.08857130
## MODE.OF.ARRIVAL_AMBULANCE                         0.38174734
## MODE.OF.ARRIVAL_TRANSFERRED                      -0.05745753
## MODE.OF.ARRIVAL_WALKED IN                        -0.34088111
##                                   LENGTH.OF.STAY..WARD IMPLANT.USED..Y.N.
## SL.                                       -0.120289533        -0.03472717
## AGE                                       -0.051822039         0.20738253
## BODY.WEIGHT                               -0.160017167         0.26406963
## BODY.HEIGHT                               -0.139424678         0.31038039
## BMI                                       -0.144448740         0.20400093
## HR.PULSE                                   0.136956581        -0.13403836
## BP..HIGH                                  -0.086199392         0.05860833
## BP.LOW                                    -0.067710833         0.08548368
## RR                                         0.199682465         0.01195301
## HB                                         0.115414190        -0.09137785
## UREA                                       0.079676229         0.21384070
## CREATININE                                -0.030626762         0.20110663
## TYPE.OF.ADMSN                              0.025982795         0.08088009
## TOTAL.COST.TO.HOSPITAL                     0.160676151         0.42327613
## TOTAL.LENGTH.OF.STAY                       0.698175009         0.13542258
## LENGTH.OF.STAY...ICU                       0.010327564         0.14663485
## LENGTH.OF.STAY..WARD                       1.000000000         0.04455770
## IMPLANT.USED..Y.N.                         0.044557699         1.00000000
## COST.OF.IMPLANT                           -0.007510964         0.80167288
```

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## KEY.COMPLAINTS..CODE_                          -0.108463135        -0.17573386
## KEY.COMPLAINTS..CODE_ACHD                      -0.044132373        -0.06678365
## KEY.COMPLAINTS..CODE_CAD-DVD                    0.061694967        -0.01088117
## KEY.COMPLAINTS..CODE_CAD-SVD                    0.152677908         0.06848321
## KEY.COMPLAINTS..CODE_CAD-TVD                   -0.107987183        -0.05966636
## KEY.COMPLAINTS..CODE_CAD-VSD                    0.048325934        -0.03157352
## KEY.COMPLAINTS..CODE_OS-ASD                     0.030243459        -0.12590390
## KEY.COMPLAINTS..CODE_other- heart               0.035146181         0.02761988
## KEY.COMPLAINTS..CODE_other- respiratory        -0.014885168        -0.12590390
## KEY.COMPLAINTS..CODE_other-general             -0.036552477        -0.03157352
## KEY.COMPLAINTS..CODE_other-nervous              0.005317226        -0.05490970
## KEY.COMPLAINTS..CODE_other-tertalogy            0.059053352        -0.13881738
## KEY.COMPLAINTS..CODE_PM-VSD                    -0.062449051        -0.07813389
## KEY.COMPLAINTS..CODE_RHD                        0.077313175         0.65660875
## PAST.MEDICAL.HISTORY.CODE_                      0.071417891         0.06214396
## PAST.MEDICAL.HISTORY.CODE_Diabetes1            -0.030254478        -0.05023453
## PAST.MEDICAL.HISTORY.CODE_Diabetes2            -0.071210199        -0.04214175
## PAST.MEDICAL.HISTORY.CODE_hypertension1         0.023450355         0.15056593
## PAST.MEDICAL.HISTORY.CODE_Hypertension1        -0.053727796        -0.05490970
## PAST.MEDICAL.HISTORY.CODE_hypertension2        -0.043403353        -0.11671032
## PAST.MEDICAL.HISTORY.CODE_hypertension3        -0.074749799        -0.07117924
## PAST.MEDICAL.HISTORY.CODE_other                 0.025730596        -0.04094018
## MODE.OF.ARRIVAL_AMBULANCE                       0.007928324         0.09542046
## MODE.OF.ARRIVAL_TRANSFERRED                     0.011845614        -0.06353406
## MODE.OF.ARRIVAL_WALKED IN                      -0.011855109        -0.06719632
##                                       COST.OF.IMPLANT KEY.COMPLAINTS..CODE_
## SL.                                      -0.099933285          -0.219849757
## AGE                                       0.182170045           0.075873824
## BODY.WEIGHT                               0.277594502           0.135766520
## BODY.HEIGHT                               0.277072494           0.168373850
## BMI                                       0.233200323           0.095346232
## HR.PULSE                                 -0.073422918          -0.137959872
## BP..HIGH                                  0.034988944           0.067992607
## BP.LOW                                    0.078585523           0.027808338
## RR                                       -0.021318370           0.028514878
## HB                                       -0.090055980          -0.078148042
## UREA                                      0.237516916          -0.111489246
## CREATININE                                0.203029200           0.151328725
## TYPE.OF.ADMSN                             0.078510830           0.012116066
## TOTAL.COST.TO.HOSPITAL                    0.483825185          -0.180963411
## TOTAL.LENGTH.OF.STAY                      0.107599429          -0.161785915
## LENGTH.OF.STAY...ICU                      0.158738379          -0.122419071
## LENGTH.OF.STAY..WARD                     -0.007510964          -0.108463135
## IMPLANT.USED..Y.N.                        0.801672884          -0.175733857
## COST.OF.IMPLANT                           1.000000000          -0.152789137
## KEY.COMPLAINTS..CODE_                    -0.152789137           1.000000000
## KEY.COMPLAINTS..CODE_ACHD                -0.071006796          -0.118697721
## KEY.COMPLAINTS..CODE_CAD-DVD              0.009539061          -0.144035341
## KEY.COMPLAINTS..CODE_CAD-SVD              0.047440994          -0.037156155
## KEY.COMPLAINTS..CODE_CAD-TVD             -0.065442706          -0.134885396
## KEY.COMPLAINTS..CODE_CAD-VSD             -0.025311636          -0.026220130
## KEY.COMPLAINTS..CODE_OS-ASD              -0.100933740          -0.104556488
## KEY.COMPLAINTS..CODE_other- heart        -0.001235068          -0.219981425
## KEY.COMPLAINTS..CODE_other- respiratory  -0.100933740          -0.104556488
## KEY.COMPLAINTS..CODE_other-general       -0.025311636          -0.026220130
## KEY.COMPLAINTS..CODE_other-nervous       -0.044019619          -0.045599587
```

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## KEY.COMPLAINTS..CODE_other-tertalogy       -0.111286128          -0.115280448
## KEY.COMPLAINTS..CODE_PM-VSD                -0.062637821          -0.064886039
## KEY.COMPLAINTS..CODE_RHD                    0.587551730          -0.141024155
## PAST.MEDICAL.HISTORY.CODE_                  0.058508650          -0.002813748
## PAST.MEDICAL.HISTORY.CODE_Diabetes1        -0.045276398          -0.084468512
## PAST.MEDICAL.HISTORY.CODE_Diabetes2        -0.039048307          -0.079966046
## PAST.MEDICAL.HISTORY.CODE_hypertension1     0.140573068           0.004068269
## PAST.MEDICAL.HISTORY.CODE_Hypertension1    -0.044019619           0.059110575
## PAST.MEDICAL.HISTORY.CODE_hypertension2    -0.093563496           0.108519179
## PAST.MEDICAL.HISTORY.CODE_hypertension3    -0.057062469           0.103771899
## PAST.MEDICAL.HISTORY.CODE_other            -0.063998029          -0.056537953
## MODE.OF.ARRIVAL_AMBULANCE                   0.090550895           0.022647065
## MODE.OF.ARRIVAL_TRANSFERRED                -0.050933530           0.128972932
## MODE.OF.ARRIVAL_WALKED IN                  -0.067194561          -0.068707546
##                                       KEY.COMPLAINTS..CODE_ACHD
## SL.                                            1.089515e-01
## AGE                                           -1.290652e-01
## BODY.WEIGHT                                   -1.571430e-01
## BODY.HEIGHT                                   -1.106088e-01
## BMI                                           -1.719210e-01
## HR.PULSE                                       1.171611e-01
## BP..HIGH                                      -8.237462e-02
## BP.LOW                                        -1.259251e-01
## RR                                             1.057372e-01
## HB                                            -9.684387e-02
## UREA                                          -5.106049e-05
## CREATININE                                    -1.471138e-01
## TYPE.OF.ADMSN                                 -6.564567e-02
## TOTAL.COST.TO.HOSPITAL                        -1.653203e-01
## TOTAL.LENGTH.OF.STAY                          -1.229474e-01
## LENGTH.OF.STAY...ICU                          -1.302475e-01
## LENGTH.OF.STAY..WARD                          -4.413237e-02
## IMPLANT.USED..Y.N.                            -6.678365e-02
## COST.OF.IMPLANT                               -7.100680e-02
## KEY.COMPLAINTS..CODE_                         -1.186977e-01
## KEY.COMPLAINTS..CODE_ACHD                      1.000000e+00
## KEY.COMPLAINTS..CODE_CAD-DVD                  -1.006804e-01
## KEY.COMPLAINTS..CODE_CAD-SVD                  -2.597207e-02
## KEY.COMPLAINTS..CODE_CAD-TVD                  -9.428458e-02
## KEY.COMPLAINTS..CODE_CAD-VSD                  -1.832781e-02
## KEY.COMPLAINTS..CODE_OS-ASD                   -7.308474e-02
## KEY.COMPLAINTS..CODE_other- heart             -1.537665e-01
## KEY.COMPLAINTS..CODE_other- respiratory       -7.308474e-02
## KEY.COMPLAINTS..CODE_other-general            -1.832781e-02
## KEY.COMPLAINTS..CODE_other-nervous            -3.187401e-02
## KEY.COMPLAINTS..CODE_other-tertalogy          -8.058077e-02
## KEY.COMPLAINTS..CODE_PM-VSD                   -4.535519e-02
## KEY.COMPLAINTS..CODE_RHD                      -9.857556e-02
## PAST.MEDICAL.HISTORY.CODE_                     2.462201e-02
## PAST.MEDICAL.HISTORY.CODE_Diabetes1           -5.904329e-02
## PAST.MEDICAL.HISTORY.CODE_Diabetes2           -5.589608e-02
## PAST.MEDICAL.HISTORY.CODE_hypertension1       -8.531131e-02
## PAST.MEDICAL.HISTORY.CODE_Hypertension1       -3.187401e-02
## PAST.MEDICAL.HISTORY.CODE_hypertension2        2.742836e-04
## PAST.MEDICAL.HISTORY.CODE_hypertension3       -4.131816e-02
## PAST.MEDICAL.HISTORY.CODE_other                1.813015e-01
```

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## MODE.OF.ARRIVAL_AMBULANCE                          -6.036323e-02
## MODE.OF.ARRIVAL_TRANSFERRED                         8.346588e-02
## MODE.OF.ARRIVAL_WALKED IN                           2.665940e-02
##                                        KEY.COMPLAINTS..CODE_CAD-DVD
## SL.                                                    -0.221423261
## AGE                                                     0.422341534
## BODY.WEIGHT                                             0.339140830
## BODY.HEIGHT                                             0.253671342
## BMI                                                     0.302586428
## HR.PULSE                                               -0.134903084
## BP..HIGH                                                0.229317505
## BP.LOW                                                  0.204021626
## RR                                                     -0.214764591
## HB                                                     -0.035178883
## UREA                                                    0.056891458
## CREATININE                                              0.235686758
## TYPE.OF.ADMSN                                           0.328862255
## TOTAL.COST.TO.HOSPITAL                                  0.318524329
## TOTAL.LENGTH.OF.STAY                                    0.277715276
## LENGTH.OF.STAY...ICU                                    0.320306143
## LENGTH.OF.STAY..WARD                                    0.061694967
## IMPLANT.USED..Y.N.                                     -0.010881167
## COST.OF.IMPLANT                                         0.009539061
## KEY.COMPLAINTS..CODE_                                  -0.144035341
## KEY.COMPLAINTS..CODE_ACHD                              -0.100680370
## KEY.COMPLAINTS..CODE_CAD-DVD                            1.000000000
## KEY.COMPLAINTS..CODE_CAD-SVD                           -0.031516152
## KEY.COMPLAINTS..CODE_CAD-TVD                           -0.114410888
## KEY.COMPLAINTS..CODE_CAD-VSD                           -0.022240127
## KEY.COMPLAINTS..CODE_OS-ASD                            -0.088685662
## KEY.COMPLAINTS..CODE_other- heart                      -0.186590030
## KEY.COMPLAINTS..CODE_other- respiratory                -0.088685662
## KEY.COMPLAINTS..CODE_other-general                     -0.022240127
## KEY.COMPLAINTS..CODE_other-nervous                     -0.038677940
## KEY.COMPLAINTS..CODE_other-tertalogy                   -0.097781812
## KEY.COMPLAINTS..CODE_PM-VSD                            -0.055036865
## KEY.COMPLAINTS..CODE_RHD                               -0.119617833
## PAST.MEDICAL.HISTORY.CODE_                             -0.361746999
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                     0.323207024
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                     0.070619119
## PAST.MEDICAL.HISTORY.CODE_hypertension1                 0.229282411
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                 0.198164752
## PAST.MEDICAL.HISTORY.CODE_hypertension2                 0.092046849
## PAST.MEDICAL.HISTORY.CODE_hypertension3                 0.041967421
## PAST.MEDICAL.HISTORY.CODE_other                        -0.088685662
## MODE.OF.ARRIVAL_AMBULANCE                               0.307030634
## MODE.OF.ARRIVAL_TRANSFERRED                            -0.044752863
## MODE.OF.ARRIVAL_WALKED IN                              -0.274697186
##                                        KEY.COMPLAINTS..CODE_CAD-SVD
## SL.                                                   -7.619830e-02
## AGE                                                    1.190274e-01
## BODY.WEIGHT                                            8.582236e-02
## BODY.HEIGHT                                            8.483050e-02
## BMI                                                    6.317185e-02
## HR.PULSE                                              -9.667147e-02
## BP..HIGH                                              -9.355195e-05
```

```
## BP.LOW                                                -4.259825e-02
## RR                                                    -3.623429e-02
## HB                                                     6.484456e-03
## UREA                                                   8.075619e-02
## CREATININE                                             6.245702e-02
## TYPE.OF.ADMSN                                         -3.470528e-02
## TOTAL.COST.TO.HOSPITAL                                 5.195805e-02
## TOTAL.LENGTH.OF.STAY                                   9.025682e-02
## LENGTH.OF.STAY...ICU                                  -2.287868e-02
## LENGTH.OF.STAY..WARD                                   1.526779e-01
## IMPLANT.USED..Y.N.                                     6.848321e-02
## COST.OF.IMPLANT                                        4.744099e-02
## KEY.COMPLAINTS..CODE_                                 -3.715615e-02
## KEY.COMPLAINTS..CODE_ACHD                             -2.597207e-02
## KEY.COMPLAINTS..CODE_CAD-DVD                          -3.151615e-02
## KEY.COMPLAINTS..CODE_CAD-SVD                           1.000000e+00
## KEY.COMPLAINTS..CODE_CAD-TVD                          -2.951407e-02
## KEY.COMPLAINTS..CODE_CAD-VSD                          -5.737186e-03
## KEY.COMPLAINTS..CODE_OS-ASD                           -2.287785e-02
## KEY.COMPLAINTS..CODE_other- heart                     -4.813380e-02
## KEY.COMPLAINTS..CODE_other- respiratory               -2.287785e-02
## KEY.COMPLAINTS..CODE_other-general                    -5.737186e-03
## KEY.COMPLAINTS..CODE_other-nervous                    -9.977576e-03
## KEY.COMPLAINTS..CODE_other-tertalogy                  -2.522434e-02
## KEY.COMPLAINTS..CODE_PM-VSD                           -1.419761e-02
## KEY.COMPLAINTS..CODE_RHD                              -3.085728e-02
## PAST.MEDICAL.HISTORY.CODE_                            -1.369430e-01
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                   -1.848243e-02
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                   -1.749725e-02
## PAST.MEDICAL.HISTORY.CODE_hypertension1                3.044387e-01
## PAST.MEDICAL.HISTORY.CODE_Hypertension1               -9.977576e-03
## PAST.MEDICAL.HISTORY.CODE_hypertension2               -2.120729e-02
## PAST.MEDICAL.HISTORY.CODE_hypertension3               -1.293389e-02
## PAST.MEDICAL.HISTORY.CODE_other                       -2.287785e-02
## MODE.OF.ARRIVAL_AMBULANCE                             -3.344874e-02
## MODE.OF.ARRIVAL_TRANSFERRED                           -1.154470e-02
## MODE.OF.ARRIVAL_WALKED IN                              3.594016e-02
##                                       KEY.COMPLAINTS..CODE_CAD-TVD
## SL.                                                    -0.111828099
## AGE                                                     0.380295880
## BODY.WEIGHT                                             0.416169194
## BODY.HEIGHT                                             0.293301004
## BMI                                                     0.351001280
## HR.PULSE                                               -0.248655543
## BP..HIGH                                                0.279842846
## BP.LOW                                                  0.248330999
## RR                                                     -0.060374717
## HB                                                     -0.005521639
## UREA                                                    0.086042330
## CREATININE                                              0.223895477
## TYPE.OF.ADMSN                                           0.158797574
## TOTAL.COST.TO.HOSPITAL                                  0.213478889
## TOTAL.LENGTH.OF.STAY                                    0.074228777
## LENGTH.OF.STAY...ICU                                    0.211297979
## LENGTH.OF.STAY..WARD                                   -0.107987183
## IMPLANT.USED..Y.N.                                     -0.059666359
```

```
## COST.OF.IMPLANT                                    -0.065442706
## KEY.COMPLAINTS..CODE_                              -0.134885396
## KEY.COMPLAINTS..CODE_ACHD                          -0.094284580
## KEY.COMPLAINTS..CODE_CAD-DVD                       -0.114410888
## KEY.COMPLAINTS..CODE_CAD-SVD                       -0.029514067
## KEY.COMPLAINTS..CODE_CAD-TVD                        1.000000000
## KEY.COMPLAINTS..CODE_CAD-VSD                       -0.020827308
## KEY.COMPLAINTS..CODE_OS-ASD                        -0.083051844
## KEY.COMPLAINTS..CODE_other- heart                  -0.174736769
## KEY.COMPLAINTS..CODE_other- respiratory            -0.083051844
## KEY.COMPLAINTS..CODE_other-general                 -0.020827308
## KEY.COMPLAINTS..CODE_other-nervous                 -0.036220897
## KEY.COMPLAINTS..CODE_other-tertalogy               -0.091570154
## KEY.COMPLAINTS..CODE_PM-VSD                        -0.051540610
## KEY.COMPLAINTS..CODE_RHD                           -0.112019028
## PAST.MEDICAL.HISTORY.CODE_                         -0.289276808
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                 0.140900465
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                 0.446985827
## PAST.MEDICAL.HISTORY.CODE_hypertension1             0.103408898
## PAST.MEDICAL.HISTORY.CODE_Hypertension1            -0.036220897
## PAST.MEDICAL.HISTORY.CODE_hypertension2             0.045402795
## PAST.MEDICAL.HISTORY.CODE_hypertension3            -0.046953014
## PAST.MEDICAL.HISTORY.CODE_other                    -0.025838352
## MODE.OF.ARRIVAL_AMBULANCE                           0.129521754
## MODE.OF.ARRIVAL_TRANSFERRED                        -0.041909907
## MODE.OF.ARRIVAL_WALKED IN                          -0.107446754
##                                     KEY.COMPLAINTS..CODE_CAD-VSD
## SL.                                                 0.026218993
## AGE                                                -0.063709098
## BODY.WEIGHT                                        -0.067656221
## BODY.HEIGHT                                        -0.060584620
## BMI                                                -0.066178815
## HR.PULSE                                            0.036951180
## BP..HIGH                                           -0.074335334
## BP.LOW                                             -0.095641672
## RR                                                 -0.025569597
## HB                                                 -0.036886498
## UREA                                                0.005807126
## CREATININE                                         -0.055804212
## TYPE.OF.ADMSN                                      -0.024490612
## TOTAL.COST.TO.HOSPITAL                             -0.030258262
## TOTAL.LENGTH.OF.STAY                                0.004576662
## LENGTH.OF.STAY...ICU                               -0.040962644
## LENGTH.OF.STAY..WARD                                0.048325934
## IMPLANT.USED..Y.N.                                 -0.031573522
## COST.OF.IMPLANT                                    -0.025311636
## KEY.COMPLAINTS..CODE_                              -0.026220130
## KEY.COMPLAINTS..CODE_ACHD                          -0.018327810
## KEY.COMPLAINTS..CODE_CAD-DVD                       -0.022240127
## KEY.COMPLAINTS..CODE_CAD-SVD                       -0.005737186
## KEY.COMPLAINTS..CODE_CAD-TVD                       -0.020827308
## KEY.COMPLAINTS..CODE_CAD-VSD                        1.000000000
## KEY.COMPLAINTS..CODE_OS-ASD                        -0.016144299
## KEY.COMPLAINTS..CODE_other- heart                  -0.033966767
## KEY.COMPLAINTS..CODE_other- respiratory            -0.016144299
## KEY.COMPLAINTS..CODE_other-general                 -0.004048583
```

```
## KEY.COMPLAINTS..CODE_other-nervous                        -0.007040915
## KEY.COMPLAINTS..CODE_other-tertalogy                      -0.017800158
## KEY.COMPLAINTS..CODE_PM-VSD                               -0.010018887
## KEY.COMPLAINTS..CODE_RHD                                  -0.021775178
## PAST.MEDICAL.HISTORY.CODE_                                 0.041894702
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                       -0.013042566
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                       -0.012347352
## PAST.MEDICAL.HISTORY.CODE_hypertension1                   -0.018845128
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                   -0.007040915
## PAST.MEDICAL.HISTORY.CODE_hypertension2                   -0.014965432
## PAST.MEDICAL.HISTORY.CODE_hypertension3                   -0.009127112
## PAST.MEDICAL.HISTORY.CODE_other                           -0.016144299
## MODE.OF.ARRIVAL_AMBULANCE                                 -0.023603907
## MODE.OF.ARRIVAL_TRANSFERRED                               -0.008146792
## MODE.OF.ARRIVAL_WALKED IN                                  0.025362035
##                                        KEY.COMPLAINTS..CODE_OS-ASD
## SL.                                                        0.22150838
## AGE                                                       -0.12119929
## BODY.WEIGHT                                               -0.06893765
## BODY.HEIGHT                                               -0.03216137
## BMI                                                       -0.01293197
## HR.PULSE                                                  -0.03626072
## BP..HIGH                                                  -0.15029982
## BP.LOW                                                    -0.07119905
## RR                                                        -0.06665814
## HB                                                        -0.04513218
## UREA                                                      -0.08179222
## CREATININE                                                -0.13291428
## TYPE.OF.ADMSN                                             -0.09765979
## TOTAL.COST.TO.HOSPITAL                                    -0.11942267
## TOTAL.LENGTH.OF.STAY                                      -0.06346976
## LENGTH.OF.STAY...ICU                                      -0.11056338
## LENGTH.OF.STAY..WARD                                       0.03024346
## IMPLANT.USED..Y.N.                                        -0.12590390
## COST.OF.IMPLANT                                           -0.10093374
## KEY.COMPLAINTS..CODE_                                     -0.10455649
## KEY.COMPLAINTS..CODE_ACHD                                 -0.07308474
## KEY.COMPLAINTS..CODE_CAD-DVD                              -0.08868566
## KEY.COMPLAINTS..CODE_CAD-SVD                              -0.02287785
## KEY.COMPLAINTS..CODE_CAD-TVD                              -0.08305184
## KEY.COMPLAINTS..CODE_CAD-VSD                              -0.01614430
## KEY.COMPLAINTS..CODE_OS-ASD                                1.00000000
## KEY.COMPLAINTS..CODE_other- heart                         -0.13544730
## KEY.COMPLAINTS..CODE_other- respiratory                   -0.06437768
## KEY.COMPLAINTS..CODE_other-general                        -0.01614430
## KEY.COMPLAINTS..CODE_other-nervous                        -0.02807665
## KEY.COMPLAINTS..CODE_other-tertalogy                      -0.07098066
## KEY.COMPLAINTS..CODE_PM-VSD                               -0.03995173
## KEY.COMPLAINTS..CODE_RHD                                  -0.08683161
## PAST.MEDICAL.HISTORY.CODE_                                 0.16706107
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                       -0.05200908
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                       -0.04923682
## PAST.MEDICAL.HISTORY.CODE_hypertension1                   -0.07514762
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                   -0.02807665
## PAST.MEDICAL.HISTORY.CODE_hypertension2                   -0.05967678
## PAST.MEDICAL.HISTORY.CODE_hypertension3                   -0.03639565
```

```
## PAST.MEDICAL.HISTORY.CODE_other                          -0.06437768
## MODE.OF.ARRIVAL_AMBULANCE                                -0.09412393
## MODE.OF.ARRIVAL_TRANSFERRED                              -0.03248649
## MODE.OF.ARRIVAL_WALKED IN                                 0.10113471
##                                        KEY.COMPLAINTS..CODE_other- heart
## SL.                                                        0.097140127
## AGE                                                       -0.210717435
## BODY.WEIGHT                                               -0.274451205
## BODY.HEIGHT                                               -0.274406076
## BMI                                                       -0.187775455
## HR.PULSE                                                   0.100259548
## BP..HIGH                                                  -0.138246026
## BP.LOW                                                    -0.114061841
## RR                                                         0.051359051
## HB                                                         0.094681176
## UREA                                                      -0.053953688
## CREATININE                                                -0.133347226
## TYPE.OF.ADMSN                                             -0.089660128
## TOTAL.COST.TO.HOSPITAL                                    -0.041382843
## TOTAL.LENGTH.OF.STAY                                       0.045611269
## LENGTH.OF.STAY...ICU                                       0.029858515
## LENGTH.OF.STAY..WARD                                       0.035146181
## IMPLANT.USED..Y.N.                                         0.027619877
## COST.OF.IMPLANT                                           -0.001235068
## KEY.COMPLAINTS..CODE_                                     -0.219981425
## KEY.COMPLAINTS..CODE_ACHD                                 -0.153766507
## KEY.COMPLAINTS..CODE_CAD-DVD                              -0.186590030
## KEY.COMPLAINTS..CODE_CAD-SVD                              -0.048133798
## KEY.COMPLAINTS..CODE_CAD-TVD                              -0.174736769
## KEY.COMPLAINTS..CODE_CAD-VSD                              -0.033966767
## KEY.COMPLAINTS..CODE_OS-ASD                               -0.135447302
## KEY.COMPLAINTS..CODE_other- heart                          1.000000000
## KEY.COMPLAINTS..CODE_other- respiratory                   -0.135447302
## KEY.COMPLAINTS..CODE_other-general                        -0.033966767
## KEY.COMPLAINTS..CODE_other-nervous                        -0.059071810
## KEY.COMPLAINTS..CODE_other-tertalogy                      -0.149339614
## KEY.COMPLAINTS..CODE_PM-VSD                               -0.084056370
## KEY.COMPLAINTS..CODE_RHD                                  -0.182689201
## PAST.MEDICAL.HISTORY.CODE_                                 0.119037206
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                       -0.060083948
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                       -0.103591708
## PAST.MEDICAL.HISTORY.CODE_hypertension1                   -0.086814950
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                   -0.059071810
## PAST.MEDICAL.HISTORY.CODE_hypertension2                   -0.082007066
## PAST.MEDICAL.HISTORY.CODE_hypertension3                    0.061538107
## PAST.MEDICAL.HISTORY.CODE_other                            0.068134097
## MODE.OF.ARRIVAL_AMBULANCE                                 -0.078972706
## MODE.OF.ARRIVAL_TRANSFERRED                               -0.068349883
## MODE.OF.ARRIVAL_WALKED IN                                  0.099905218
##                                        KEY.COMPLAINTS..CODE_other- respiratory
## SL.                                                                 -0.09722741
## AGE                                                                 -0.19278765
## BODY.WEIGHT                                                         -0.22434071
## BODY.HEIGHT                                                         -0.20134836
## BMI                                                                 -0.21848758
## HR.PULSE                                                             0.22579910
```

```
## BP..HIGH                                                -0.05551125
## BP.LOW                                                  -0.07599283
## RR                                                       0.13634134
## HB                                                      -0.03080295
## UREA                                                    -0.04868265
## CREATININE                                              -0.06069906
## TYPE.OF.ADMSN                                           -0.09765979
## TOTAL.COST.TO.HOSPITAL                                  -0.09428871
## TOTAL.LENGTH.OF.STAY                                    -0.04146826
## LENGTH.OF.STAY...ICU                                    -0.04458716
## LENGTH.OF.STAY..WARD                                    -0.01488517
## IMPLANT.USED..Y.N.                                      -0.12590390
## COST.OF.IMPLANT                                         -0.10093374
## KEY.COMPLAINTS..CODE_                                   -0.10455649
## KEY.COMPLAINTS..CODE_ACHD                               -0.07308474
## KEY.COMPLAINTS..CODE_CAD-DVD                            -0.08868566
## KEY.COMPLAINTS..CODE_CAD-SVD                            -0.02287785
## KEY.COMPLAINTS..CODE_CAD-TVD                            -0.08305184
## KEY.COMPLAINTS..CODE_CAD-VSD                            -0.01614430
## KEY.COMPLAINTS..CODE_OS-ASD                             -0.06437768
## KEY.COMPLAINTS..CODE_other- heart                       -0.13544730
## KEY.COMPLAINTS..CODE_other- respiratory                  1.00000000
## KEY.COMPLAINTS..CODE_other-general                      -0.01614430
## KEY.COMPLAINTS..CODE_other-nervous                      -0.02807665
## KEY.COMPLAINTS..CODE_other-tertalogy                    -0.07098066
## KEY.COMPLAINTS..CODE_PM-VSD                             -0.03995173
## KEY.COMPLAINTS..CODE_RHD                                -0.08683161
## PAST.MEDICAL.HISTORY.CODE_                               0.13023338
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                     -0.05200908
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                     -0.04923682
## PAST.MEDICAL.HISTORY.CODE_hypertension1                 -0.07514762
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                 -0.02807665
## PAST.MEDICAL.HISTORY.CODE_hypertension2                  0.01621984
## PAST.MEDICAL.HISTORY.CODE_hypertension3                 -0.03639565
## PAST.MEDICAL.HISTORY.CODE_other                         -0.06437768
## MODE.OF.ARRIVAL_AMBULANCE                               -0.09412393
## MODE.OF.ARRIVAL_TRANSFERRED                              0.10179099
## MODE.OF.ARRIVAL_WALKED IN                                0.05195548
##                                     KEY.COMPLAINTS..CODE_other-general
## SL.                                                      0.030662890
## AGE                                                      0.111073520
## BODY.WEIGHT                                              0.072970875
## BODY.HEIGHT                                              0.040330678
## BMI                                                      0.073542029
## HR.PULSE                                                 0.030672391
## BP..HIGH                                                 0.044495573
## BP.LOW                                                  -0.008200078
## RR                                                       0.007630718
## HB                                                      -0.065910184
## UREA                                                     0.345645077
## CREATININE                                               0.143952802
## TYPE.OF.ADMSN                                            0.165311631
## TOTAL.COST.TO.HOSPITAL                                  -0.074289145
## TOTAL.LENGTH.OF.STAY                                    -0.042715514
## LENGTH.OF.STAY...ICU                                    -0.024417472
## LENGTH.OF.STAY..WARD                                    -0.036552477
```

```
## IMPLANT.USED..Y.N.                                    -0.031573522
## COST.OF.IMPLANT                                       -0.025311636
## KEY.COMPLAINTS..CODE_                                 -0.026220130
## KEY.COMPLAINTS..CODE_ACHD                             -0.018327810
## KEY.COMPLAINTS..CODE_CAD-DVD                          -0.022240127
## KEY.COMPLAINTS..CODE_CAD-SVD                          -0.005737186
## KEY.COMPLAINTS..CODE_CAD-TVD                          -0.020827308
## KEY.COMPLAINTS..CODE_CAD-VSD                          -0.004048583
## KEY.COMPLAINTS..CODE_OS-ASD                           -0.016144299
## KEY.COMPLAINTS..CODE_other- heart                     -0.033966767
## KEY.COMPLAINTS..CODE_other- respiratory               -0.016144299
## KEY.COMPLAINTS..CODE_other-general                     1.000000000
## KEY.COMPLAINTS..CODE_other-nervous                    -0.007040915
## KEY.COMPLAINTS..CODE_other-tertalogy                  -0.017800158
## KEY.COMPLAINTS..CODE_PM-VSD                           -0.010018887
## KEY.COMPLAINTS..CODE_RHD                              -0.021775178
## PAST.MEDICAL.HISTORY.CODE_                            -0.096637112
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                   -0.013042566
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                   -0.012347352
## PAST.MEDICAL.HISTORY.CODE_hypertension1               -0.018845128
## PAST.MEDICAL.HISTORY.CODE_Hypertension1               -0.007040915
## PAST.MEDICAL.HISTORY.CODE_hypertension2               -0.014965432
## PAST.MEDICAL.HISTORY.CODE_hypertension3               -0.009127112
## PAST.MEDICAL.HISTORY.CODE_other                        0.250774778
## MODE.OF.ARRIVAL_AMBULANCE                              0.171521728
## MODE.OF.ARRIVAL_TRANSFERRED                           -0.008146792
## MODE.OF.ARRIVAL_WALKED IN                             -0.159631631
##                                     KEY.COMPLAINTS..CODE_other-nervous
## SL.                                                   -0.006955568
## AGE                                                   -0.117932218
## BODY.WEIGHT                                           -0.143236811
## BODY.HEIGHT                                           -0.166694564
## BMI                                                   -0.138247698
## HR.PULSE                                               0.038783234
## BP..HIGH                                              -0.068963057
## BP.LOW                                                -0.060197271
## RR                                                     0.013270627
## HB                                                    -0.080974620
## UREA                                                   0.038581821
## CREATININE                                            -0.064699628
## TYPE.OF.ADMSN                                         -0.042591771
## TOTAL.COST.TO.HOSPITAL                                -0.036708394
## TOTAL.LENGTH.OF.STAY                                   0.014813143
## LENGTH.OF.STAY...ICU                                   0.015083045
## LENGTH.OF.STAY..WARD                                   0.005317226
## IMPLANT.USED..Y.N.                                    -0.054909702
## COST.OF.IMPLANT                                       -0.044019619
## KEY.COMPLAINTS..CODE_                                 -0.045599587
## KEY.COMPLAINTS..CODE_ACHD                             -0.031874006
## KEY.COMPLAINTS..CODE_CAD-DVD                          -0.038677940
## KEY.COMPLAINTS..CODE_CAD-SVD                          -0.009977576
## KEY.COMPLAINTS..CODE_CAD-TVD                          -0.036220897
## KEY.COMPLAINTS..CODE_CAD-VSD                          -0.007040915
## KEY.COMPLAINTS..CODE_OS-ASD                           -0.028076648
## KEY.COMPLAINTS..CODE_other- heart                     -0.059071810
## KEY.COMPLAINTS..CODE_other- respiratory               -0.028076648
```

```
## KEY.COMPLAINTS..CODE_other-general                              -0.007040915
## KEY.COMPLAINTS..CODE_other-nervous                               1.000000000
## KEY.COMPLAINTS..CODE_other-tertalogy                            -0.030956362
## KEY.COMPLAINTS..CODE_PM-VSD                                     -0.017423906
## KEY.COMPLAINTS..CODE_RHD                                        -0.037869343
## PAST.MEDICAL.HISTORY.CODE_                                       0.072859329
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                             -0.022682405
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                             -0.021473354
## PAST.MEDICAL.HISTORY.CODE_hypertension1                         -0.032773676
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                         -0.012244898
## PAST.MEDICAL.HISTORY.CODE_hypertension2                         -0.026026474
## PAST.MEDICAL.HISTORY.CODE_hypertension3                         -0.015873016
## PAST.MEDICAL.HISTORY.CODE_other                                 -0.028076648
## MODE.OF.ARRIVAL_AMBULANCE                                       -0.041049698
## MODE.OF.ARRIVAL_TRANSFERRED                                     -0.014168134
## MODE.OF.ARRIVAL_WALKED IN                                        0.044107268
##                                       KEY.COMPLAINTS..CODE_other-tertalogy
## SL.                                                            0.143497311
## AGE                                                           -0.247082741
## BODY.WEIGHT                                                   -0.242231767
## BODY.HEIGHT                                                   -0.182083862
## BMI                                                           -0.224644235
## HR.PULSE                                                       0.167828751
## BP..HIGH                                                      -0.096718973
## BP.LOW                                                        -0.037785179
## RR                                                             0.009221211
## HB                                                             0.403951405
## UREA                                                          -0.068313855
## CREATININE                                                    -0.172347077
## TYPE.OF.ADMSN                                                 -0.107676380
## TOTAL.COST.TO.HOSPITAL                                        -0.070798836
## TOTAL.LENGTH.OF.STAY                                          -0.037635465
## LENGTH.OF.STAY...ICU                                          -0.107354811
## LENGTH.OF.STAY..WARD                                           0.059053352
## IMPLANT.USED..Y.N.                                            -0.138817378
## COST.OF.IMPLANT                                               -0.111286128
## KEY.COMPLAINTS..CODE_                                         -0.115280448
## KEY.COMPLAINTS..CODE_ACHD                                     -0.080580766
## KEY.COMPLAINTS..CODE_CAD-DVD                                  -0.097781812
## KEY.COMPLAINTS..CODE_CAD-SVD                                  -0.025224338
## KEY.COMPLAINTS..CODE_CAD-TVD                                  -0.091570154
## KEY.COMPLAINTS..CODE_CAD-VSD                                  -0.017800158
## KEY.COMPLAINTS..CODE_OS-ASD                                   -0.070980655
## KEY.COMPLAINTS..CODE_other- heart                             -0.149339614
## KEY.COMPLAINTS..CODE_other- respiratory                       -0.070980655
## KEY.COMPLAINTS..CODE_other-general                            -0.017800158
## KEY.COMPLAINTS..CODE_other-nervous                            -0.030956362
## KEY.COMPLAINTS..CODE_other-tertalogy                           1.000000000
## KEY.COMPLAINTS..CODE_PM-VSD                                   -0.044049428
## KEY.COMPLAINTS..CODE_RHD                                      -0.095737597
## PAST.MEDICAL.HISTORY.CODE_                                     0.082683484
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                           -0.057343455
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                           -0.054286850
## PAST.MEDICAL.HISTORY.CODE_hypertension1                       -0.082855226
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                       -0.030956362
## PAST.MEDICAL.HISTORY.CODE_hypertension2                        0.003936609
```

```
## PAST.MEDICAL.HISTORY.CODE_hypertension3                        -0.040128618
## PAST.MEDICAL.HISTORY.CODE_other                                 0.059413437
## MODE.OF.ARRIVAL_AMBULANCE                                      -0.103777861
## MODE.OF.ARRIVAL_TRANSFERRED                                    -0.035818502
## MODE.OF.ARRIVAL_WALKED IN                                       0.111507712
##                                        KEY.COMPLAINTS..CODE_PM-VSD
## SL.                                                     0.161291634
## AGE                                                    -0.149617303
## BODY.WEIGHT                                            -0.186760091
## BODY.HEIGHT                                            -0.272778456
## BMI                                                    -0.090687303
## HR.PULSE                                                0.158772692
## BP..HIGH                                               -0.147142259
## BP.LOW                                                 -0.103690236
## RR                                                      0.005190193
## HB                                                     -0.052804635
## UREA                                                   -0.022781261
## CREATININE                                             -0.127798155
## TYPE.OF.ADMSN                                          -0.060606061
## TOTAL.COST.TO.HOSPITAL                                 -0.069586023
## TOTAL.LENGTH.OF.STAY                                   -0.086201206
## LENGTH.OF.STAY...ICU                                   -0.060425064
## LENGTH.OF.STAY..WARD                                   -0.062449051
## IMPLANT.USED..Y.N.                                     -0.078133890
## COST.OF.IMPLANT                                        -0.062637821
## KEY.COMPLAINTS..CODE_                                  -0.064886039
## KEY.COMPLAINTS..CODE_ACHD                              -0.045355191
## KEY.COMPLAINTS..CODE_CAD-DVD                           -0.055036865
## KEY.COMPLAINTS..CODE_CAD-SVD                           -0.014197615
## KEY.COMPLAINTS..CODE_CAD-TVD                           -0.051540610
## KEY.COMPLAINTS..CODE_CAD-VSD                           -0.010018887
## KEY.COMPLAINTS..CODE_OS-ASD                            -0.039951732
## KEY.COMPLAINTS..CODE_other- heart                      -0.084056370
## KEY.COMPLAINTS..CODE_other- respiratory                -0.039951732
## KEY.COMPLAINTS..CODE_other-general                     -0.010018887
## KEY.COMPLAINTS..CODE_other-nervous                     -0.017423906
## KEY.COMPLAINTS..CODE_other-tertalogy                   -0.044049428
## KEY.COMPLAINTS..CODE_PM-VSD                             1.000000000
## KEY.COMPLAINTS..CODE_RHD                               -0.053886271
## PAST.MEDICAL.HISTORY.CODE_                              0.103675353
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                    -0.032275982
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                    -0.030555560
## PAST.MEDICAL.HISTORY.CODE_hypertension1                -0.046635380
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                -0.017423906
## PAST.MEDICAL.HISTORY.CODE_hypertension2                -0.037034432
## PAST.MEDICAL.HISTORY.CODE_hypertension3                -0.022586545
## PAST.MEDICAL.HISTORY.CODE_other                        -0.039951732
## MODE.OF.ARRIVAL_AMBULANCE                              -0.058411764
## MODE.OF.ARRIVAL_TRANSFERRED                            -0.020160580
## MODE.OF.ARRIVAL_WALKED IN                               0.062762540
##                                        KEY.COMPLAINTS..CODE_RHD
## SL.                                                0.073358798
## AGE                                                0.040293455
## BODY.WEIGHT                                        0.158034685
## BODY.HEIGHT                                        0.245200590
## BMI                                                0.093054032
```

```
## HR.PULSE                                          -0.070771974
## BP..HIGH                                           0.024254101
## BP.LOW                                             0.038878290
## RR                                                 0.027305665
## HB                                                -0.121627946
## UREA                                               0.121291359
## CREATININE                                         0.026174967
## TYPE.OF.ADMSN                                     -0.053195421
## TOTAL.COST.TO.HOSPITAL                             0.158573732
## TOTAL.LENGTH.OF.STAY                               0.009940849
## LENGTH.OF.STAY...ICU                              -0.059453938
## LENGTH.OF.STAY..WARD                               0.077313175
## IMPLANT.USED..Y.N.                                 0.656608753
## COST.OF.IMPLANT                                    0.587551730
## KEY.COMPLAINTS..CODE_                             -0.141024155
## KEY.COMPLAINTS..CODE_ACHD                         -0.098575558
## KEY.COMPLAINTS..CODE_CAD-DVD                      -0.119617833
## KEY.COMPLAINTS..CODE_CAD-SVD                      -0.030857279
## KEY.COMPLAINTS..CODE_CAD-TVD                      -0.112019028
## KEY.COMPLAINTS..CODE_CAD-VSD                      -0.021775178
## KEY.COMPLAINTS..CODE_OS-ASD                       -0.086831610
## KEY.COMPLAINTS..CODE_other- heart                 -0.182689201
## KEY.COMPLAINTS..CODE_other- respiratory           -0.086831610
## KEY.COMPLAINTS..CODE_other-general                -0.021775178
## KEY.COMPLAINTS..CODE_other-nervous                -0.037869343
## KEY.COMPLAINTS..CODE_other-tertalogy              -0.095737597
## KEY.COMPLAINTS..CODE_PM-VSD                       -0.053886271
## KEY.COMPLAINTS..CODE_RHD                           1.000000000
## PAST.MEDICAL.HISTORY.CODE_                         0.139357537
## PAST.MEDICAL.HISTORY.CODE_Diabetes1               -0.070149036
## PAST.MEDICAL.HISTORY.CODE_Diabetes2               -0.066409849
## PAST.MEDICAL.HISTORY.CODE_hypertension1           -0.004678059
## PAST.MEDICAL.HISTORY.CODE_Hypertension1           -0.037869343
## PAST.MEDICAL.HISTORY.CODE_hypertension2           -0.080491113
## PAST.MEDICAL.HISTORY.CODE_hypertension3           -0.049089889
## PAST.MEDICAL.HISTORY.CODE_other                   -0.031615612
## MODE.OF.ARRIVAL_AMBULANCE                         -0.046223870
## MODE.OF.ARRIVAL_TRANSFERRED                       -0.043817265
## MODE.OF.ARRIVAL_WALKED IN                          0.059871787
##                                     PAST.MEDICAL.HISTORY.CODE_
## SL.                                                0.207547481
## AGE                                               -0.495499218
## BODY.WEIGHT                                       -0.422591719
## BODY.HEIGHT                                       -0.267999581
## BMI                                               -0.377508333
## HR.PULSE                                           0.224909127
## BP..HIGH                                          -0.305092607
## BP.LOW                                            -0.164882840
## RR                                                 0.168398133
## HB                                                 0.084911135
## UREA                                              -0.111422835
## CREATININE                                        -0.345523482
## TYPE.OF.ADMSN                                     -0.217948631
## TOTAL.COST.TO.HOSPITAL                            -0.171344291
## TOTAL.LENGTH.OF.STAY                              -0.088140847
## LENGTH.OF.STAY...ICU                              -0.187906809
```

```
## LENGTH.OF.STAY..WARD                                  0.071417891
## IMPLANT.USED..Y.N.                                    0.062143959
## COST.OF.IMPLANT                                       0.058508650
## KEY.COMPLAINTS..CODE_                                -0.002813748
## KEY.COMPLAINTS..CODE_ACHD                             0.024622010
## KEY.COMPLAINTS..CODE_CAD-DVD                         -0.361746999
## KEY.COMPLAINTS..CODE_CAD-SVD                         -0.136943008
## KEY.COMPLAINTS..CODE_CAD-TVD                         -0.289276808
## KEY.COMPLAINTS..CODE_CAD-VSD                          0.041894702
## KEY.COMPLAINTS..CODE_OS-ASD                           0.167061066
## KEY.COMPLAINTS..CODE_other- heart                     0.119037206
## KEY.COMPLAINTS..CODE_other- respiratory               0.130233382
## KEY.COMPLAINTS..CODE_other-general                   -0.096637112
## KEY.COMPLAINTS..CODE_other-nervous                    0.072859329
## KEY.COMPLAINTS..CODE_other-tertalogy                  0.082683484
## KEY.COMPLAINTS..CODE_PM-VSD                           0.103675353
## KEY.COMPLAINTS..CODE_RHD                              0.139357537
## PAST.MEDICAL.HISTORY.CODE_                            1.000000000
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                  -0.311317792
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                  -0.294723472
## PAST.MEDICAL.HISTORY.CODE_hypertension1              -0.449821277
## PAST.MEDICAL.HISTORY.CODE_Hypertension1              -0.168062185
## PAST.MEDICAL.HISTORY.CODE_hypertension2              -0.357215393
## PAST.MEDICAL.HISTORY.CODE_hypertension3              -0.217858388
## PAST.MEDICAL.HISTORY.CODE_other                      -0.385354193
## MODE.OF.ARRIVAL_AMBULANCE                            -0.240344982
## MODE.OF.ARRIVAL_TRANSFERRED                          -0.055077913
## MODE.OF.ARRIVAL_WALKED IN                             0.248037311
##                                      PAST.MEDICAL.HISTORY.CODE_Diabetes1
## SL.                                                       -0.008303324
## AGE                                                        0.232523176
## BODY.WEIGHT                                                0.234188384
## BODY.HEIGHT                                                0.148802613
## BMI                                                        0.216367222
## HR.PULSE                                                  -0.073119718
## BP..HIGH                                                   0.191195030
## BP.LOW                                                     0.170769345
## RR                                                        -0.173284794
## HB                                                        -0.011305123
## UREA                                                      -0.085495420
## CREATININE                                                 0.059413356
## TYPE.OF.ADMSN                                              0.165683374
## TOTAL.COST.TO.HOSPITAL                                     0.087828392
## TOTAL.LENGTH.OF.STAY                                       0.060449500
## LENGTH.OF.STAY...ICU                                       0.113220581
## LENGTH.OF.STAY..WARD                                      -0.030254478
## IMPLANT.USED..Y.N.                                        -0.050234526
## COST.OF.IMPLANT                                           -0.045276398
## KEY.COMPLAINTS..CODE_                                     -0.084468512
## KEY.COMPLAINTS..CODE_ACHD                                 -0.059043294
## KEY.COMPLAINTS..CODE_CAD-DVD                               0.323207024
## KEY.COMPLAINTS..CODE_CAD-SVD                              -0.018482426
## KEY.COMPLAINTS..CODE_CAD-TVD                               0.140900465
## KEY.COMPLAINTS..CODE_CAD-VSD                              -0.013042566
## KEY.COMPLAINTS..CODE_OS-ASD                               -0.052009082
## KEY.COMPLAINTS..CODE_other- heart                         -0.060083948
```

```
## KEY.COMPLAINTS..CODE_other- respiratory                          -0.052009082
## KEY.COMPLAINTS..CODE_other-general                               -0.013042566
## KEY.COMPLAINTS..CODE_other-nervous                               -0.022682405
## KEY.COMPLAINTS..CODE_other-tertalogy                             -0.057343455
## KEY.COMPLAINTS..CODE_PM-VSD                                      -0.032275982
## KEY.COMPLAINTS..CODE_RHD                                         -0.070149036
## PAST.MEDICAL.HISTORY.CODE_                                       -0.311317792
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                               1.000000000
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                              -0.039777165
## PAST.MEDICAL.HISTORY.CODE_hypertension1                          -0.060709841
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                          -0.022682405
## PAST.MEDICAL.HISTORY.CODE_hypertension2                          -0.048211347
## PAST.MEDICAL.HISTORY.CODE_hypertension3                          -0.029403118
## PAST.MEDICAL.HISTORY.CODE_other                                  -0.052009082
## MODE.OF.ARRIVAL_AMBULANCE                                         0.175399658
## MODE.OF.ARRIVAL_TRANSFERRED                                      -0.026245002
## MODE.OF.ARRIVAL_WALKED IN                                        -0.156679715
##                                      PAST.MEDICAL.HISTORY.CODE_Diabetes2
## SL.                                                           -0.23988774
## AGE                                                            0.21097775
## BODY.WEIGHT                                                    0.24497158
## BODY.HEIGHT                                                    0.18918767
## BMI                                                            0.19438366
## HR.PULSE                                                      -0.04369015
## BP..HIGH                                                       0.15583631
## BP.LOW                                                         0.07869994
## RR                                                            -0.02735496
## HB                                                            -0.07245319
## UREA                                                           0.18143143
## CREATININE                                                     0.26979941
## TYPE.OF.ADMSN                                                  0.18257890
## TOTAL.COST.TO.HOSPITAL                                         0.30959079
## TOTAL.LENGTH.OF.STAY                                           0.13815722
## LENGTH.OF.STAY...ICU                                           0.26753429
## LENGTH.OF.STAY..WARD                                          -0.07121020
## IMPLANT.USED..Y.N.                                            -0.04214175
## COST.OF.IMPLANT                                               -0.03904831
## KEY.COMPLAINTS..CODE_                                         -0.07996605
## KEY.COMPLAINTS..CODE_ACHD                                     -0.05589608
## KEY.COMPLAINTS..CODE_CAD-DVD                                   0.07061912
## KEY.COMPLAINTS..CODE_CAD-SVD                                  -0.01749725
## KEY.COMPLAINTS..CODE_CAD-TVD                                   0.44698583
## KEY.COMPLAINTS..CODE_CAD-VSD                                  -0.01234735
## KEY.COMPLAINTS..CODE_OS-ASD                                   -0.04923682
## KEY.COMPLAINTS..CODE_other- heart                             -0.10359171
## KEY.COMPLAINTS..CODE_other- respiratory                       -0.04923682
## KEY.COMPLAINTS..CODE_other-general                            -0.01234735
## KEY.COMPLAINTS..CODE_other-nervous                            -0.02147335
## KEY.COMPLAINTS..CODE_other-tertalogy                          -0.05428685
## KEY.COMPLAINTS..CODE_PM-VSD                                   -0.03055556
## KEY.COMPLAINTS..CODE_RHD                                      -0.06640985
## PAST.MEDICAL.HISTORY.CODE_                                    -0.29472347
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                           -0.03977716
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                            1.00000000
## PAST.MEDICAL.HISTORY.CODE_hypertension1                       -0.05747380
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                       -0.02147335
```

```
## PAST.MEDICAL.HISTORY.CODE_hypertension2                          -0.04564151
## PAST.MEDICAL.HISTORY.CODE_hypertension3                          -0.02783583
## PAST.MEDICAL.HISTORY.CODE_other                                  -0.04923682
## MODE.OF.ARRIVAL_AMBULANCE                                         0.19249884
## MODE.OF.ARRIVAL_TRANSFERRED                                      -0.02484605
## MODE.OF.ARRIVAL_WALKED IN                                        -0.17340338
##                                        PAST.MEDICAL.HISTORY.CODE_hypertension1
## SL.                                                            -0.164447464
## AGE                                                             0.378368640
## BODY.WEIGHT                                                     0.283187068
## BODY.HEIGHT                                                     0.233944969
## BMI                                                             0.233928908
## HR.PULSE                                                       -0.236437380
## BP..HIGH                                                        0.193287427
## BP.LOW                                                          0.155164485
## RR                                                             -0.088112188
## HB                                                              0.003881730
## UREA                                                            0.075898442
## CREATININE                                                      0.306677861
## TYPE.OF.ADMSN                                                   0.151046813
## TOTAL.COST.TO.HOSPITAL                                          0.184849101
## TOTAL.LENGTH.OF.STAY                                            0.103853116
## LENGTH.OF.STAY...ICU                                            0.121234290
## LENGTH.OF.STAY..WARD                                            0.023450355
## IMPLANT.USED..Y.N.                                              0.150565931
## COST.OF.IMPLANT                                                 0.140573068
## KEY.COMPLAINTS..CODE_                                           0.004068269
## KEY.COMPLAINTS..CODE_ACHD                                      -0.085311314
## KEY.COMPLAINTS..CODE_CAD-DVD                                    0.229282411
## KEY.COMPLAINTS..CODE_CAD-SVD                                    0.304438708
## KEY.COMPLAINTS..CODE_CAD-TVD                                    0.103408898
## KEY.COMPLAINTS..CODE_CAD-VSD                                   -0.018845128
## KEY.COMPLAINTS..CODE_OS-ASD                                    -0.075147622
## KEY.COMPLAINTS..CODE_other- heart                              -0.086814950
## KEY.COMPLAINTS..CODE_other- respiratory                        -0.075147622
## KEY.COMPLAINTS..CODE_other-general                             -0.018845128
## KEY.COMPLAINTS..CODE_other-nervous                             -0.032773676
## KEY.COMPLAINTS..CODE_other-tertalogy                           -0.082855226
## KEY.COMPLAINTS..CODE_PM-VSD                                    -0.046635380
## KEY.COMPLAINTS..CODE_RHD                                       -0.004678059
## PAST.MEDICAL.HISTORY.CODE_                                     -0.449821277
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                            -0.060709841
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                            -0.057473796
## PAST.MEDICAL.HISTORY.CODE_hypertension1                         1.000000000
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                        -0.032773676
## PAST.MEDICAL.HISTORY.CODE_hypertension2                        -0.069660297
## PAST.MEDICAL.HISTORY.CODE_hypertension3                        -0.042484395
## PAST.MEDICAL.HISTORY.CODE_other                                -0.075147622
## MODE.OF.ARRIVAL_AMBULANCE                                       0.162607911
## MODE.OF.ARRIVAL_TRANSFERRED                                    -0.037921251
## MODE.OF.ARRIVAL_WALKED IN                                      -0.140275748
##                                        PAST.MEDICAL.HISTORY.CODE_Hypertension1
## SL.                                                             0.013653521
## AGE                                                             0.137512973
## BODY.WEIGHT                                                     0.104525553
## BODY.HEIGHT                                                     0.108825340
```

```
## BMI                                                          0.073103267
## HR.PULSE                                                    -0.084970873
## BP..HIGH                                                     0.077382520
## BP.LOW                                                      -0.014260806
## RR                                                          -0.005975662
## HB                                                          -0.052131609
## UREA                                                         0.005352105
## CREATININE                                                   0.054937458
## TYPE.OF.ADMSN                                               -0.042591771
## TOTAL.COST.TO.HOSPITAL                                      -0.021327816
## TOTAL.LENGTH.OF.STAY                                        -0.019456069
## LENGTH.OF.STAY...ICU                                         0.024674315
## LENGTH.OF.STAY..WARD                                        -0.053727796
## IMPLANT.USED..Y.N.                                          -0.054909702
## COST.OF.IMPLANT                                             -0.044019619
## KEY.COMPLAINTS..CODE_                                        0.059110575
## KEY.COMPLAINTS..CODE_ACHD                                   -0.031874006
## KEY.COMPLAINTS..CODE_CAD-DVD                                 0.198164752
## KEY.COMPLAINTS..CODE_CAD-SVD                                -0.009977576
## KEY.COMPLAINTS..CODE_CAD-TVD                                -0.036220897
## KEY.COMPLAINTS..CODE_CAD-VSD                                -0.007040915
## KEY.COMPLAINTS..CODE_OS-ASD                                 -0.028076648
## KEY.COMPLAINTS..CODE_other- heart                           -0.059071810
## KEY.COMPLAINTS..CODE_other- respiratory                     -0.028076648
## KEY.COMPLAINTS..CODE_other-general                          -0.007040915
## KEY.COMPLAINTS..CODE_other-nervous                          -0.012244898
## KEY.COMPLAINTS..CODE_other-tertalogy                        -0.030956362
## KEY.COMPLAINTS..CODE_PM-VSD                                 -0.017423906
## KEY.COMPLAINTS..CODE_RHD                                    -0.037869343
## PAST.MEDICAL.HISTORY.CODE_                                  -0.168062185
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                         -0.022682405
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                         -0.021473354
## PAST.MEDICAL.HISTORY.CODE_hypertension1                     -0.032773676
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                      1.000000000
## PAST.MEDICAL.HISTORY.CODE_hypertension2                     -0.026026474
## PAST.MEDICAL.HISTORY.CODE_hypertension3                     -0.015873016
## PAST.MEDICAL.HISTORY.CODE_other                             -0.028076648
## MODE.OF.ARRIVAL_AMBULANCE                                   -0.041049698
## MODE.OF.ARRIVAL_TRANSFERRED                                 -0.014168134
## MODE.OF.ARRIVAL_WALKED IN                                    0.044107268
##                                      PAST.MEDICAL.HISTORY.CODE_hypertension2
## SL.                                                         -0.0372759642
## AGE                                                          0.1501881843
## BODY.WEIGHT                                                  0.1670239889
## BODY.HEIGHT                                                  0.1014107828
## BMI                                                          0.1559424786
## HR.PULSE                                                    -0.0740804412
## BP..HIGH                                                     0.1222402334
## BP.LOW                                                       0.0194155714
## RR                                                          -0.0850767558
## HB                                                          -0.0286519459
## UREA                                                        -0.0139479024
## CREATININE                                                   0.1146723505
## TYPE.OF.ADMSN                                               -0.0365596315
## TOTAL.COST.TO.HOSPITAL                                      -0.0938240274
## TOTAL.LENGTH.OF.STAY                                        -0.0704894040
```

```
## LENGTH.OF.STAY...ICU                                    -0.0667357150
## LENGTH.OF.STAY..WARD                                    -0.0434033533
## IMPLANT.USED..Y.N.                                      -0.1167103165
## COST.OF.IMPLANT                                         -0.0935634961
## KEY.COMPLAINTS..CODE_                                    0.1085191788
## KEY.COMPLAINTS..CODE_ACHD                                0.0002742836
## KEY.COMPLAINTS..CODE_CAD-DVD                             0.0920468486
## KEY.COMPLAINTS..CODE_CAD-SVD                            -0.0212072907
## KEY.COMPLAINTS..CODE_CAD-TVD                             0.0454027947
## KEY.COMPLAINTS..CODE_CAD-VSD                            -0.0149654324
## KEY.COMPLAINTS..CODE_OS-ASD                             -0.0596767844
## KEY.COMPLAINTS..CODE_other- heart                       -0.0820070660
## KEY.COMPLAINTS..CODE_other- respiratory                  0.0162198440
## KEY.COMPLAINTS..CODE_other-general                      -0.0149654324
## KEY.COMPLAINTS..CODE_other-nervous                      -0.0260264737
## KEY.COMPLAINTS..CODE_other-tertalogy                     0.0039366088
## KEY.COMPLAINTS..CODE_PM-VSD                             -0.0370344319
## KEY.COMPLAINTS..CODE_RHD                                -0.0804911128
## PAST.MEDICAL.HISTORY.CODE_                              -0.3572153928
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                     -0.0482113471
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                     -0.0456415147
## PAST.MEDICAL.HISTORY.CODE_hypertension1                 -0.0696602966
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                 -0.0260264737
## PAST.MEDICAL.HISTORY.CODE_hypertension2                  1.0000000000
## PAST.MEDICAL.HISTORY.CODE_hypertension3                 -0.0337380215
## PAST.MEDICAL.HISTORY.CODE_other                         -0.0596767844
## MODE.OF.ARRIVAL_AMBULANCE                               -0.0317682918
## MODE.OF.ARRIVAL_TRANSFERRED                              0.2571298248
## MODE.OF.ARRIVAL_WALKED IN                               -0.0640552883
##                                      PAST.MEDICAL.HISTORY.CODE_hypertension3
## SL.                                                        -0.169710123
## AGE                                                         0.179367500
## BODY.WEIGHT                                                 0.135910505
## BODY.HEIGHT                                                 0.079913133
## BMI                                                         0.139851212
## HR.PULSE                                                   -0.147421949
## BP..HIGH                                                    0.140494475
## BP.LOW                                                      0.080077791
## RR                                                         -0.132490697
## HB                                                         -0.008378623
## UREA                                                       -0.033060818
## CREATININE                                                  0.054327789
## TYPE.OF.ADMSN                                               0.115944265
## TOTAL.COST.TO.HOSPITAL                                      0.049971669
## TOTAL.LENGTH.OF.STAY                                        0.020979145
## LENGTH.OF.STAY...ICU                                        0.101610736
## LENGTH.OF.STAY..WARD                                       -0.074749799
## IMPLANT.USED..Y.N.                                         -0.071179243
## COST.OF.IMPLANT                                            -0.057062469
## KEY.COMPLAINTS..CODE_                                       0.103771899
## KEY.COMPLAINTS..CODE_ACHD                                  -0.041318155
## KEY.COMPLAINTS..CODE_CAD-DVD                                0.041967421
## KEY.COMPLAINTS..CODE_CAD-SVD                               -0.012933894
## KEY.COMPLAINTS..CODE_CAD-TVD                               -0.046953014
## KEY.COMPLAINTS..CODE_CAD-VSD                               -0.009127112
## KEY.COMPLAINTS..CODE_OS-ASD                                -0.036395655
```

```
## KEY.COMPLAINTS..CODE_other- heart                                  0.061538107
## KEY.COMPLAINTS..CODE_other- respiratory                           -0.036395655
## KEY.COMPLAINTS..CODE_other-general                                -0.009127112
## KEY.COMPLAINTS..CODE_other-nervous                                -0.015873016
## KEY.COMPLAINTS..CODE_other-tertalogy                              -0.040128618
## KEY.COMPLAINTS..CODE_PM-VSD                                       -0.022586545
## KEY.COMPLAINTS..CODE_RHD                                          -0.049089889
## PAST.MEDICAL.HISTORY.CODE_                                        -0.217858388
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                               -0.029403118
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                               -0.027835830
## PAST.MEDICAL.HISTORY.CODE_hypertension1                           -0.042484395
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                           -0.015873016
## PAST.MEDICAL.HISTORY.CODE_hypertension2                           -0.033738021
## PAST.MEDICAL.HISTORY.CODE_hypertension3                            1.000000000
## PAST.MEDICAL.HISTORY.CODE_other                                   -0.036395655
## MODE.OF.ARRIVAL_AMBULANCE                                          0.122743665
## MODE.OF.ARRIVAL_TRANSFERRED                                       -0.018366100
## MODE.OF.ARRIVAL_WALKED IN                                         -0.109643557
##                                      PAST.MEDICAL.HISTORY.CODE_other
## SL.                                                       0.11164022
## AGE                                                      -0.14410443
## BODY.WEIGHT                                              -0.17889264
## BODY.HEIGHT                                              -0.21389676
## BMI                                                      -0.13265357
## HR.PULSE                                                  0.12648344
## BP..HIGH                                                 -0.14557882
## BP.LOW                                                   -0.12103270
## RR                                                        0.10103708
## HB                                                       -0.04623443
## UREA                                                      0.08628832
## CREATININE                                               -0.10968129
## TYPE.OF.ADMSN                                            -0.04720223
## TOTAL.COST.TO.HOSPITAL                                   -0.12831817
## TOTAL.LENGTH.OF.STAY                                     -0.04461133
## LENGTH.OF.STAY...ICU                                     -0.08857130
## LENGTH.OF.STAY..WARD                                      0.02573060
## IMPLANT.USED..Y.N.                                       -0.04094018
## COST.OF.IMPLANT                                          -0.06399803
## KEY.COMPLAINTS..CODE_                                    -0.05653795
## KEY.COMPLAINTS..CODE_ACHD                                 0.18130145
## KEY.COMPLAINTS..CODE_CAD-DVD                             -0.08868566
## KEY.COMPLAINTS..CODE_CAD-SVD                             -0.02287785
## KEY.COMPLAINTS..CODE_CAD-TVD                             -0.02583835
## KEY.COMPLAINTS..CODE_CAD-VSD                             -0.01614430
## KEY.COMPLAINTS..CODE_OS-ASD                              -0.06437768
## KEY.COMPLAINTS..CODE_other- heart                         0.06813410
## KEY.COMPLAINTS..CODE_other- respiratory                  -0.06437768
## KEY.COMPLAINTS..CODE_other-general                        0.25077478
## KEY.COMPLAINTS..CODE_other-nervous                       -0.02807665
## KEY.COMPLAINTS..CODE_other-tertalogy                      0.05941344
## KEY.COMPLAINTS..CODE_PM-VSD                              -0.03995173
## KEY.COMPLAINTS..CODE_RHD                                 -0.03161561
## PAST.MEDICAL.HISTORY.CODE_                               -0.38535419
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                      -0.05200908
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                      -0.04923682
## PAST.MEDICAL.HISTORY.CODE_hypertension1                  -0.07514762
```

```
## PAST.MEDICAL.HISTORY.CODE_Hypertension1                    -0.02807665
## PAST.MEDICAL.HISTORY.CODE_hypertension2                    -0.05967678
## PAST.MEDICAL.HISTORY.CODE_hypertension3                    -0.03639565
## PAST.MEDICAL.HISTORY.CODE_other                             1.00000000
## MODE.OF.ARRIVAL_AMBULANCE                                  -0.04225119
## MODE.OF.ARRIVAL_TRANSFERRED                                -0.03248649
## MODE.OF.ARRIVAL_WALKED IN                                   0.05195548
##                                        MODE.OF.ARRIVAL_AMBULANCE
## SL.                                                -0.242505005
## AGE                                                 0.449037162
## BODY.WEIGHT                                         0.353625841
## BODY.HEIGHT                                         0.264552158
## BMI                                                 0.318102954
## HR.PULSE                                           -0.118296657
## BP..HIGH                                            0.221889667
## BP.LOW                                              0.179902378
## RR                                                  0.018679901
## HB                                                 -0.054745307
## UREA                                                0.413538946
## CREATININE                                          0.471613210
## TYPE.OF.ADMSN                                       0.963794108
## TOTAL.COST.TO.HOSPITAL                              0.353738721
## TOTAL.LENGTH.OF.STAY                                0.277129450
## LENGTH.OF.STAY...ICU                                0.381747345
## LENGTH.OF.STAY..WARD                                0.007928324
## IMPLANT.USED..Y.N.                                  0.095420463
## COST.OF.IMPLANT                                     0.090550895
## KEY.COMPLAINTS..CODE_                               0.022647065
## KEY.COMPLAINTS..CODE_ACHD                          -0.060363228
## KEY.COMPLAINTS..CODE_CAD-DVD                        0.307030634
## KEY.COMPLAINTS..CODE_CAD-SVD                       -0.033448745
## KEY.COMPLAINTS..CODE_CAD-TVD                        0.129521754
## KEY.COMPLAINTS..CODE_CAD-VSD                       -0.023603907
## KEY.COMPLAINTS..CODE_OS-ASD                        -0.094123929
## KEY.COMPLAINTS..CODE_other- heart                  -0.078972706
## KEY.COMPLAINTS..CODE_other- respiratory            -0.094123929
## KEY.COMPLAINTS..CODE_other-general                  0.171521728
## KEY.COMPLAINTS..CODE_other-nervous                 -0.041049698
## KEY.COMPLAINTS..CODE_other-tertalogy               -0.103777861
## KEY.COMPLAINTS..CODE_PM-VSD                        -0.058411764
## KEY.COMPLAINTS..CODE_RHD                           -0.046223870
## PAST.MEDICAL.HISTORY.CODE_                         -0.240344982
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                 0.175399658
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                 0.192498841
## PAST.MEDICAL.HISTORY.CODE_hypertension1             0.162607911
## PAST.MEDICAL.HISTORY.CODE_Hypertension1            -0.041049698
## PAST.MEDICAL.HISTORY.CODE_hypertension2            -0.031768292
## PAST.MEDICAL.HISTORY.CODE_hypertension3             0.122743665
## PAST.MEDICAL.HISTORY.CODE_other                    -0.042251186
## MODE.OF.ARRIVAL_AMBULANCE                           1.000000000
## MODE.OF.ARRIVAL_TRANSFERRED                        -0.047497140
## MODE.OF.ARRIVAL_WALKED IN                          -0.930678773
##                                        MODE.OF.ARRIVAL_TRANSFERRED
## SL.                                                    0.001788453
## AGE                                                    0.030317007
## BODY.WEIGHT                                            0.024767281
```

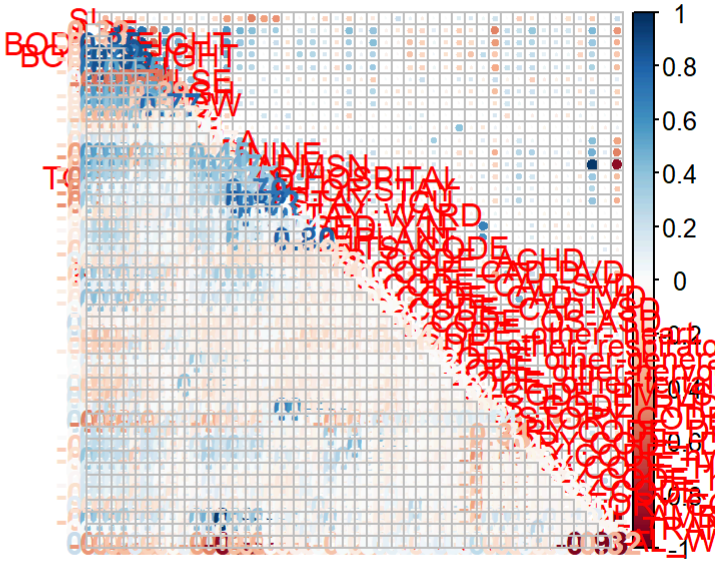
```
## BODY.HEIGHT                                          0.016468864
## BMI                                                  0.022400907
## HR.PULSE                                            -0.004610588
## BP..HIGH                                             0.044701854
## BP.LOW                                              -0.038495045
## RR                                                   0.032056865
## HB                                                  -0.082568437
## UREA                                                 0.019924487
## CREATININE                                           0.120092155
## TYPE.OF.ADMSN                                       -0.049281418
## TOTAL.COST.TO.HOSPITAL                              -0.075391827
## TOTAL.LENGTH.OF.STAY                                -0.032424843
## LENGTH.OF.STAY...ICU                                -0.057457529
## LENGTH.OF.STAY..WARD                                 0.011845614
## IMPLANT.USED..Y.N.                                  -0.063534056
## COST.OF.IMPLANT                                     -0.050933530
## KEY.COMPLAINTS..CODE_                                0.128972932
## KEY.COMPLAINTS..CODE_ACHD                            0.083465884
## KEY.COMPLAINTS..CODE_CAD-DVD                        -0.044752863
## KEY.COMPLAINTS..CODE_CAD-SVD                        -0.011544697
## KEY.COMPLAINTS..CODE_CAD-TVD                        -0.041909907
## KEY.COMPLAINTS..CODE_CAD-VSD                        -0.008146792
## KEY.COMPLAINTS..CODE_OS-ASD                         -0.032486487
## KEY.COMPLAINTS..CODE_other- heart                   -0.068349883
## KEY.COMPLAINTS..CODE_other- respiratory              0.101790993
## KEY.COMPLAINTS..CODE_other-general                  -0.008146792
## KEY.COMPLAINTS..CODE_other-nervous                  -0.014168134
## KEY.COMPLAINTS..CODE_other-tertalogy                -0.035818502
## KEY.COMPLAINTS..CODE_PM-VSD                         -0.020160580
## KEY.COMPLAINTS..CODE_RHD                            -0.043817265
## PAST.MEDICAL.HISTORY.CODE_                          -0.055077913
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                 -0.026245002
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                 -0.024846052
## PAST.MEDICAL.HISTORY.CODE_hypertension1             -0.037921251
## PAST.MEDICAL.HISTORY.CODE_Hypertension1             -0.014168134
## PAST.MEDICAL.HISTORY.CODE_hypertension2              0.257129825
## PAST.MEDICAL.HISTORY.CODE_hypertension3             -0.018366100
## PAST.MEDICAL.HISTORY.CODE_other                     -0.032486487
## MODE.OF.ARRIVAL_AMBULANCE                           -0.047497140
## MODE.OF.ARRIVAL_TRANSFERRED                          1.000000000
## MODE.OF.ARRIVAL_WALKED IN                           -0.321219950
##                                      MODE.OF.ARRIVAL_WALKED IN
## SL.                                                 0.22925782
## AGE                                                -0.43682437
## BODY.WEIGHT                                        -0.34433473
## BODY.HEIGHT                                        -0.25684693
## BMI                                                -0.30978969
## HR.PULSE                                            0.11384269
## BP..HIGH                                           -0.22674005
## BP.LOW                                             -0.15646204
## RR                                                 -0.02945079
## HB                                                  0.08214339
## UREA                                               -0.39936315
## CREATININE                                         -0.49110837
## TYPE.OF.ADMSN                                      -0.89569939
## TOTAL.COST.TO.HOSPITAL                             -0.30775839
```

```
## TOTAL.LENGTH.OF.STAY                                -0.25086378
## LENGTH.OF.STAY...ICU                                -0.34088111
## LENGTH.OF.STAY..WARD                                -0.01185511
## IMPLANT.USED..Y.N.                                  -0.06719632
## COST.OF.IMPLANT                                     -0.06719456
## KEY.COMPLAINTS..CODE_                               -0.06870755
## KEY.COMPLAINTS..CODE_ACHD                            0.02665940
## KEY.COMPLAINTS..CODE_CAD-DVD                        -0.27469719
## KEY.COMPLAINTS..CODE_CAD-SVD                         0.03594016
## KEY.COMPLAINTS..CODE_CAD-TVD                        -0.10744675
## KEY.COMPLAINTS..CODE_CAD-VSD                         0.02536203
## KEY.COMPLAINTS..CODE_OS-ASD                          0.10113471
## KEY.COMPLAINTS..CODE_other- heart                    0.09990522
## KEY.COMPLAINTS..CODE_other- respiratory              0.05195548
## KEY.COMPLAINTS..CODE_other-general                  -0.15963163
## KEY.COMPLAINTS..CODE_other-nervous                   0.04410727
## KEY.COMPLAINTS..CODE_other-tertalogy                 0.11150771
## KEY.COMPLAINTS..CODE_PM-VSD                          0.06276254
## KEY.COMPLAINTS..CODE_RHD                             0.05987179
## PAST.MEDICAL.HISTORY.CODE_                           0.24803731
## PAST.MEDICAL.HISTORY.CODE_Diabetes1                 -0.15667971
## PAST.MEDICAL.HISTORY.CODE_Diabetes2                 -0.17340338
## PAST.MEDICAL.HISTORY.CODE_hypertension1             -0.14027575
## PAST.MEDICAL.HISTORY.CODE_Hypertension1              0.04410727
## PAST.MEDICAL.HISTORY.CODE_hypertension2             -0.06405529
## PAST.MEDICAL.HISTORY.CODE_hypertension3             -0.10964356
## PAST.MEDICAL.HISTORY.CODE_other                      0.05195548
## MODE.OF.ARRIVAL_AMBULANCE                           -0.93067877
## MODE.OF.ARRIVAL_TRANSFERRED                         -0.32121995
## MODE.OF.ARRIVAL_WALKED IN                            1.00000000
```

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
corrplot.mixed(cr)
```



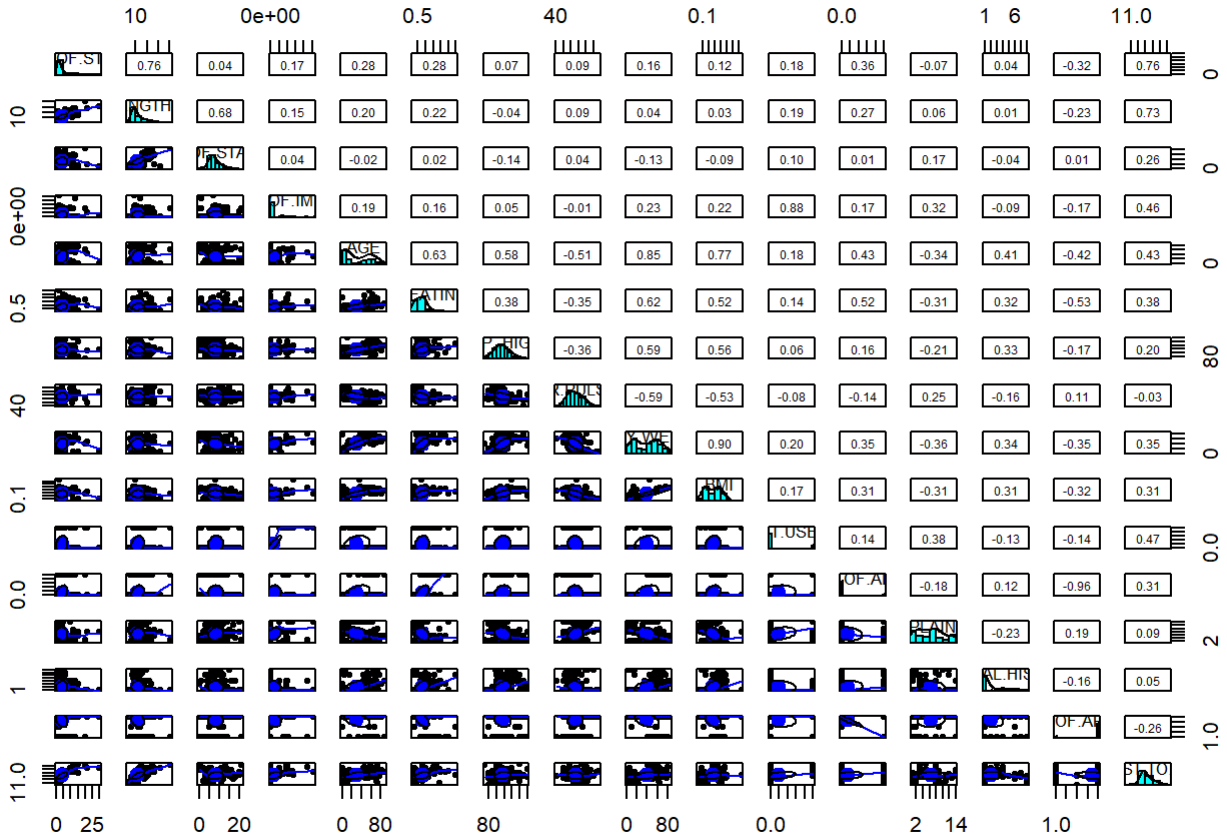

```
#divide the data set into training and testing
library(caTools)
set.seed(11)
split=sample.split(missionML$TOTAL.COST.TO.HOSPITAL,SplitRatio = 0.7)
training = subset(missionML,split==TRUE)
testing = subset(missionML,split==FALSE)

library(psych)
```

```
##
## Attaching package: 'psych'
##
## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

```
#Heteroscedasticity
training$TOTAL.COST.TO.HOSPITAL = log(training$TOTAL.COST.TO.HOSPITAL)
testing$TOTAL.COST.TO.HOSPITAL = log(testing$TOTAL.COST.TO.HOSPITAL)

pairs.panels(training[c("LENGTH.OF.STAY...ICU",
                        "TOTAL.LENGTH.OF.STAY",
                        "LENGTH.OF.STAY..WARD",
                        "COST.OF.IMPLANT",
                        "AGE",
                        "CREATININE",
                        "BP..HIGH",
                        "HR.PULSE",
                        "BODY.WEIGHT",
                        "BMI",
                        "IMPLANT.USED..Y.N.",
                        "TYPE.OF.ADMSN",
                        "KEY.COMPLAINTS..CODE",
                        "PAST.MEDICAL.HISTORY.CODE",
                        "MODE.OF.ARRIVAL","TOTAL.COST.TO.HOSPITAL")])
```




```
#Built The Model
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:psych':
##
##     logit
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
```

```
model1 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
                        TOTAL.LENGTH.OF.STAY,data = training)
summary(model1)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.08059 -0.10727  0.00502  0.14548  0.84304
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)           11.469275   0.060773 188.722  < 2e-16 ***
## LENGTH.OF.STAY...ICU   0.059783   0.008891   6.724 2.57e-10 ***
## TOTAL.LENGTH.OF.STAY   0.032462   0.006541   4.963 1.68e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3196 on 170 degrees of freedom
## Multiple R-squared:  0.6277, Adjusted R-squared:  0.6234
## F-statistic: 143.3 on 2 and 170 DF,  p-value: < 2.2e-16
```

```
vif(model1)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY
##              2.37617              2.37617
```

```
model2 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
              TOTAL.LENGTH.OF.STAY+LENGTH.OF.STAY..WARD
                        ,data = training)
summary(model2)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + LENGTH.OF.STAY..WARD, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.08282 -0.10781  0.00544  0.14616  0.84089
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          11.47072    0.06104 187.918   <2e-16 ***
## LENGTH.OF.STAY...ICU  0.09980    0.10393   0.960    0.338
## TOTAL.LENGTH.OF.STAY -0.00751    0.10363  -0.072    0.942
## LENGTH.OF.STAY..WARD  0.03965    0.10260   0.386    0.700
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3204 on 169 degrees of freedom
## Multiple R-squared:  0.6281, Adjusted R-squared:  0.6215
## F-statistic: 95.12 on 3 and 169 DF,  p-value: < 2.2e-16
```

```
vif(model2)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY LENGTH.OF.STAY..WARD
##             323.0434             593.4427             250.1012
```

```
model3 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
              TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT
            ,data = training)
summary(model3)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.93256 -0.05682  0.04228  0.14404  0.98723
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)           1.143e+01  5.137e-02 222.510  < 2e-16 ***
## LENGTH.OF.STAY...ICU 5.422e-02  7.515e-03   7.214 1.74e-11 ***
## TOTAL.LENGTH.OF.STAY 3.097e-02  5.510e-03   5.621 7.68e-08 ***
## COST.OF.IMPLANT      1.005e-05  1.195e-06   8.414 1.61e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2691 on 169 degrees of freedom
## Multiple R-squared:  0.7376, Adjusted R-squared:  0.733
## F-statistic: 158.4 on 3 and 169 DF,  p-value: < 2.2e-16
```

```
vif(model3)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.394738             2.378617             1.031273
```

```
model4 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
              TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+AGE
            ,data = training)
summary(model4)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + AGE, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.03320 -0.07478  0.02613  0.13734  0.89779
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)          1.134e+01  5.172e-02 219.267  < 2e-16 ***
## LENGTH.OF.STAY...ICU 4.738e-02  7.192e-03   6.588 5.48e-10 ***
## TOTAL.LENGTH.OF.STAY 3.168e-02  5.174e-03   6.122 6.31e-09 ***
## COST.OF.IMPLANT      9.247e-06  1.134e-06   8.157 7.68e-14 ***
## AGE                  3.735e-03  7.653e-04   4.881 2.44e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2526 on 168 degrees of freedom
## Multiple R-squared:  0.7702, Adjusted R-squared:  0.7648
## F-statistic: 140.8 on 4 and 168 DF,  p-value: < 2.2e-16
```

```
vif(model4)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.489139             2.380454             1.053694
##                  AGE
##             1.112438
```

```
model5 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
              TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+AGE+CREATININE
            ,data = training)
summary(model5)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + AGE + CREATININE,
##     data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.06859 -0.07727  0.02447  0.14791  0.88281
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)           1.132e+01  5.564e-02 203.478  < 2e-16 ***
## LENGTH.OF.STAY...ICU 4.700e-02  7.207e-03   6.522 7.91e-10 ***
## TOTAL.LENGTH.OF.STAY 3.151e-02  5.180e-03   6.084 7.76e-09 ***
## COST.OF.IMPLANT      9.205e-06  1.135e-06   8.109 1.04e-13 ***
## AGE                  3.222e-03  9.493e-04   3.394 0.000861 ***
## CREATININE           4.958e-02  5.415e-02   0.915 0.361274
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2527 on 167 degrees of freedom
## Multiple R-squared:  0.7714, Adjusted R-squared:  0.7645
## F-statistic: 112.7 on 5 and 167 DF,  p-value: < 2.2e-16
```

```
vif(model5)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.497315             2.383296             1.055394
##                  AGE           CREATININE
##             1.709711             1.687579
```

```
model6 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
              TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+AGE+BP..HIGH
            ,data = training)
summary(model6)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + AGE + BP..HIGH,
##     data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.01143 -0.07844  0.04028  0.12283  0.84914
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)           1.110e+01  1.334e-01  83.267  < 2e-16 ***
## LENGTH.OF.STAY...ICU 4.692e-02  7.139e-03   6.571 6.08e-10 ***
## TOTAL.LENGTH.OF.STAY 3.317e-02  5.192e-03   6.388 1.61e-09 ***
## COST.OF.IMPLANT      9.365e-06  1.126e-06   8.314 3.10e-14 ***
## AGE                  2.672e-03  9.409e-04   2.840  0.00507 **
## BP..HIGH             2.165e-03  1.131e-03   1.914  0.05737 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2506 on 167 degrees of freedom
## Multiple R-squared:  0.7752, Adjusted R-squared:  0.7684
## F-statistic: 115.1 on 5 and 167 DF,  p-value: < 2.2e-16
```

```
vif(model6)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.492020             2.435312             1.056889
##                  AGE             BP..HIGH
##             1.707990             1.574497
```

```
model7 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
              TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+AGE+BP..HIGH+HR.PULSE
            ,data = training)
summary(model7)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + AGE + BP..HIGH +
##     HR.PULSE, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.01790 -0.07786  0.03776  0.12060  0.84420
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)          1.108e+01  1.814e-01  61.099  < 2e-16 ***
## LENGTH.OF.STAY...ICU 4.669e-02  7.270e-03   6.423 1.35e-09 ***
## TOTAL.LENGTH.OF.STAY 3.316e-02  5.208e-03   6.366 1.82e-09 ***
## COST.OF.IMPLANT      9.352e-06  1.132e-06   8.261 4.35e-14 ***
## AGE                  2.760e-03  1.064e-03   2.594   0.0103 *
## BP..HIGH             2.176e-03  1.136e-03   1.915   0.0572 .
## HR.PULSE             2.055e-04  1.151e-03   0.178   0.8586
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2513 on 166 degrees of freedom
## Multiple R-squared:  0.7752, Adjusted R-squared:  0.7671
## F-statistic:  95.4 on 6 and 166 DF,  p-value: < 2.2e-16
```

```
vif(model7)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.568834             2.435751             1.061177
##                  AGE             BP..HIGH             HR.PULSE
##             2.170492             1.579195             1.485832
```

```
model8 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
              TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+AGE+BP..HIGH+HR.PULSE+BODY.WEIGHT
            ,data = training)
summary(model8)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + AGE + BP..HIGH +
##     HR.PULSE + BODY.WEIGHT, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.01388 -0.08458  0.02971  0.14543  0.87743
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)          1.095e+01  1.878e-01  58.295  < 2e-16 ***
## LENGTH.OF.STAY...ICU 4.480e-02  7.221e-03   6.204 4.27e-09 ***
## TOTAL.LENGTH.OF.STAY 3.582e-02  5.266e-03   6.802 1.81e-10 ***
## COST.OF.IMPLANT      8.767e-06  1.145e-06   7.655 1.54e-12 ***
## AGE                  2.809e-04  1.496e-03   0.188   0.8513
## BP..HIGH             1.683e-03  1.141e-03   1.474   0.1423
## HR.PULSE             1.130e-03  1.204e-03   0.939   0.3492
## BODY.WEIGHT          4.125e-03  1.773e-03   2.326   0.0212 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2481 on 165 degrees of freedom
## Multiple R-squared:  0.7823, Adjusted R-squared:  0.7731
## F-statistic: 84.72 on 7 and 165 DF,  p-value: < 2.2e-16
```

```
vif(model8)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.601773             2.556538             1.115042
##                  AGE             BP..HIGH             HR.PULSE
##             4.406272             1.635687             1.667658
##          BODY.WEIGHT
##             4.820608
```

```
model9 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
              TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+HR.PULSE+BODY.WEIGHT
            ,data = training)
summary(model9)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE + BODY.WEIGHT,
##     data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.00963 -0.09526  0.03128  0.14704  0.93290
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)           1.111e+01  1.478e-01  75.193  < 2e-16 ***
## LENGTH.OF.STAY...ICU 4.526e-02  7.213e-03   6.275 2.90e-09 ***
## TOTAL.LENGTH.OF.STAY 3.551e-02  5.156e-03   6.887 1.11e-10 ***
## COST.OF.IMPLANT      8.576e-06  1.140e-06   7.525 3.10e-12 ***
## HR.PULSE             1.106e-03  1.202e-03   0.920    0.359
## BODY.WEIGHT          5.307e-03  1.086e-03   4.885 2.41e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2484 on 167 degrees of freedom
## Multiple R-squared:  0.7791, Adjusted R-squared:  0.7725
## F-statistic: 117.8 on 5 and 167 DF,  p-value: < 2.2e-16
```

```
vif(model9)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.589681             2.445015             1.101142
##             HR.PULSE          BODY.WEIGHT
##             1.659316             1.804640
```

```
model10 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
              TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+HR.PULSE+BODY.WEIGHT+BMI
            ,data = training)
summary(model10)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE + BODY.WEIGHT +
##     BMI, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.00764 -0.09777  0.03154  0.14535  0.93239
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)           1.112e+01  1.502e-01  74.027  < 2e-16 ***
## LENGTH.OF.STAY...ICU  4.503e-02  7.270e-03   6.194 4.46e-09 ***
## TOTAL.LENGTH.OF.STAY  3.563e-02  5.183e-03   6.873 1.21e-10 ***
## COST.OF.IMPLANT       8.587e-06  1.143e-06   7.511 3.43e-12 ***
## HR.PULSE              1.114e-03  1.206e-03   0.924  0.35702
## BODY.WEIGHT           5.851e-03  2.014e-03   2.906  0.00416 **
## BMI                  -1.108e-01  3.448e-01  -0.321  0.74829
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.249 on 166 degrees of freedom
## Multiple R-squared:  0.7793, Adjusted R-squared:  0.7713
## F-statistic: 97.67 on 6 and 166 DF,  p-value: < 2.2e-16
```

```
vif(model10)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.616149             2.457438             1.102026
##             HR.PULSE          BODY.WEIGHT                  BMI
##             1.659911             6.166645             5.247866
```

```
model11 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
               TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+HR.PULSE+BODY.WEIGHT+IMPLANT.USED..Y.N.
             ,data = training)
summary(model11)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE + BODY.WEIGHT +
##     IMPLANT.USED..Y.N., data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.01605 -0.08143  0.03151  0.13987  0.92844
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)           1.107e+01  1.494e-01  74.053  < 2e-16 ***
## LENGTH.OF.STAY...ICU 4.557e-02  7.174e-03   6.351 1.97e-09 ***
## TOTAL.LENGTH.OF.STAY 3.443e-02  5.166e-03   6.664 3.76e-10 ***
## COST.OF.IMPLANT      4.955e-06  2.404e-06   2.062   0.0408 *
## HR.PULSE             1.550e-03  1.224e-03   1.267   0.2069
## BODY.WEIGHT          5.550e-03  1.090e-03   5.094 9.44e-07 ***
## IMPLANT.USED..Y.N.   1.780e-01  1.042e-01   1.708   0.0895 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.247 on 166 degrees of freedom
## Multiple R-squared:  0.7829, Adjusted R-squared:  0.7751
## F-statistic: 99.79 on 6 and 166 DF,  p-value: < 2.2e-16
```

```
vif(model11)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.591263             2.482439             4.954256
##             HR.PULSE          BODY.WEIGHT   IMPLANT.USED..Y.N.
##             1.737734             1.836098             4.863809
```

```
model12 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
               TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+HR.PULSE+BODY.WEIGHT+TYPE.OF.ADMSN
             ,data = training)
summary(model12)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE + BODY.WEIGHT +
##     TYPE.OF.ADMSN, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.91914 -0.10597  0.02545  0.16258  0.87810
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)           1.109e+01  1.471e-01  75.394  < 2e-16 ***
## LENGTH.OF.STAY...ICU  4.775e-02  7.290e-03   6.550 6.90e-10 ***
## TOTAL.LENGTH.OF.STAY  3.582e-02  5.123e-03   6.991 6.33e-11 ***
## COST.OF.IMPLANT       8.683e-06  1.133e-06   7.662 1.44e-12 ***
## HR.PULSE              1.130e-03  1.194e-03   0.947   0.3452
## BODY.WEIGHT           5.802e-03  1.112e-03   5.217 5.36e-07 ***
## TYPE.OF.ADMSN        -1.126e-01  6.139e-02  -1.834   0.0684 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2466 on 166 degrees of freedom
## Multiple R-squared:  0.7835, Adjusted R-squared:  0.7757
## F-statistic: 100.1 on 6 and 166 DF,  p-value: < 2.2e-16
```

```
vif(model12)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.682743             2.447625             1.104033
##             HR.PULSE          BODY.WEIGHT        TYPE.OF.ADMSN
##             1.659514             1.917688             1.280630
```

```
model13 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
               TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+HR.PULSE+BODY.WEIGHT+KEY.COMPLAINTS..CODE
             ,data = training)
summary(model13)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE + BODY.WEIGHT +
##     KEY.COMPLAINTS..CODE, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.66444 -0.10827  0.01482  0.13556  0.82660
##
## Coefficients:
##                                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)                          1.109e+01  1.497e-01  74.083  < 2e-16
## LENGTH.OF.STAY...ICU                 4.304e-02  6.952e-03   6.191 5.13e-09
## TOTAL.LENGTH.OF.STAY                 3.221e-02  4.976e-03   6.472 1.20e-09
## COST.OF.IMPLANT                      8.037e-06  1.377e-06   5.835 3.04e-08
## HR.PULSE                             1.351e-03  1.139e-03   1.186  0.23731
## BODY.WEIGHT                          4.422e-03  1.380e-03   3.204  0.00164
## KEY.COMPLAINTS..CODEACHD            -1.382e-01  8.681e-02  -1.592  0.11333
## KEY.COMPLAINTS..CODECAD-DVD          1.704e-01  7.065e-02   2.412  0.01703
## KEY.COMPLAINTS..CODECAD-SVD          1.134e-01  1.716e-01   0.661  0.50976
## KEY.COMPLAINTS..CODECAD-TVD          2.360e-01  7.284e-02   3.240  0.00146
## KEY.COMPLAINTS..CODEOS-ASD           4.996e-02  8.536e-02   0.585  0.55919
## KEY.COMPLAINTS..CODEother- heart     4.868e-02  6.690e-02   0.728  0.46789
## KEY.COMPLAINTS..CODEother- respiratory  4.006e-02  8.428e-02   0.475  0.63520
## KEY.COMPLAINTS..CODEother-general   -9.269e-01  2.298e-01  -4.034 8.60e-05
## KEY.COMPLAINTS..CODEother-nervous   -4.017e-02  1.707e-01  -0.235  0.81424
## KEY.COMPLAINTS..CODEother-tertalogy  2.622e-01  1.042e-01   2.516  0.01290
## KEY.COMPLAINTS..CODEPM-VSD           2.204e-01  1.122e-01   1.964  0.05128
## KEY.COMPLAINTS..CODERHD              1.377e-01  8.724e-02   1.579  0.11648
##
## (Intercept)                         ***
## LENGTH.OF.STAY...ICU                ***
## TOTAL.LENGTH.OF.STAY                ***
## COST.OF.IMPLANT                     ***
## HR.PULSE
## BODY.WEIGHT                         **
## KEY.COMPLAINTS..CODEACHD
## KEY.COMPLAINTS..CODECAD-DVD         *
## KEY.COMPLAINTS..CODECAD-SVD
## KEY.COMPLAINTS..CODECAD-TVD         **
## KEY.COMPLAINTS..CODEOS-ASD
## KEY.COMPLAINTS..CODEother- heart
## KEY.COMPLAINTS..CODEother- respiratory
## KEY.COMPLAINTS..CODEother-general   ***
## KEY.COMPLAINTS..CODEother-nervous
## KEY.COMPLAINTS..CODEother-tertalogy  *
## KEY.COMPLAINTS..CODEPM-VSD           .
## KEY.COMPLAINTS..CODERHD
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.223 on 155 degrees of freedom
## Multiple R-squared:  0.8347, Adjusted R-squared:  0.8166
## F-statistic: 46.06 on 17 and 155 DF,  p-value: < 2.2e-16
```

```
vif(model13)
```

```
##                           GVIF Df GVIF^(1/(2*Df))
## LENGTH.OF.STAY...ICU 2.984175  1        1.727476
## TOTAL.LENGTH.OF.STAY 2.824702  1        1.680685
## COST.OF.IMPLANT      1.995340  1        1.412565
## HR.PULSE             1.847427  1        1.359201
## BODY.WEIGHT          3.612569  1        1.900676
## KEY.COMPLAINTS..CODE 7.480789 12        1.087463
```

```
model14 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
               TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+HR.PULSE+BODY.WEIGHT+PAST.MEDICAL.HISTORY.
CODE
             ,data = training)
summary(model14)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE + BODY.WEIGHT +
##     PAST.MEDICAL.HISTORY.CODE, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.83796 -0.08780  0.03346  0.13651  0.70621
##
## Coefficients:
##                                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)                         1.117e+01  1.463e-01  76.319  < 2e-16
## LENGTH.OF.STAY...ICU                3.827e-02  7.140e-03   5.359 2.87e-07
## TOTAL.LENGTH.OF.STAY                3.686e-02  5.002e-03   7.369 8.64e-12
## COST.OF.IMPLANT                     9.240e-06  1.156e-06   7.996 2.42e-13
## HR.PULSE                            1.041e-03  1.199e-03   0.868  0.38657
## BODY.WEIGHT                         3.385e-03  1.239e-03   2.731  0.00701
## PAST.MEDICAL.HISTORY.CODEDiabetes1  1.777e-01  1.020e-01   1.742  0.08338
## PAST.MEDICAL.HISTORY.CODEDiabetes2  3.808e-01  1.044e-01   3.646  0.00036
## PAST.MEDICAL.HISTORY.CODEhypertension1  1.122e-01  6.952e-02   1.614  0.10854
## PAST.MEDICAL.HISTORY.CODEHypertension1  1.583e-01  1.731e-01   0.914  0.36195
## PAST.MEDICAL.HISTORY.CODEhypertension2  5.738e-02  8.162e-02   0.703  0.48306
## PAST.MEDICAL.HISTORY.CODEhypertension3  3.079e-02  1.267e-01   0.243  0.80825
## PAST.MEDICAL.HISTORY.CODEother         -2.375e-01  1.115e-01  -2.130  0.03472
##
## (Intercept)                         ***
## LENGTH.OF.STAY...ICU                ***
## TOTAL.LENGTH.OF.STAY                ***
## COST.OF.IMPLANT                     ***
## HR.PULSE
## BODY.WEIGHT                         **
## PAST.MEDICAL.HISTORY.CODEDiabetes1  .
## PAST.MEDICAL.HISTORY.CODEDiabetes2  ***
## PAST.MEDICAL.HISTORY.CODEhypertension1
## PAST.MEDICAL.HISTORY.CODEHypertension1
## PAST.MEDICAL.HISTORY.CODEhypertension2
## PAST.MEDICAL.HISTORY.CODEhypertension3
## PAST.MEDICAL.HISTORY.CODEother         *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.239 on 160 degrees of freedom
## Multiple R-squared:  0.804,  Adjusted R-squared:  0.7893
## F-statistic: 54.69 on 12 and 160 DF,  p-value: < 2.2e-16
```

```
vif(model14)
```

```
##                               GVIF Df GVIF^(1/(2*Df))
## LENGTH.OF.STAY...ICU      2.739222  1        1.655060
## TOTAL.LENGTH.OF.STAY      2.484013  1        1.576075
## COST.OF.IMPLANT           1.222170  1        1.105518
## HR.PULSE                  1.781135  1        1.334592
## BODY.WEIGHT               2.534584  1        1.592038
## PAST.MEDICAL.HISTORY.CODE 2.059763  7        1.052969
```

```
model15 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
               TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+HR.PULSE+BODY.WEIGHT+MODE.OF.ARRIVAL
             ,data = training)
summary(model15)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE + BODY.WEIGHT +
##     MODE.OF.ARRIVAL, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.92097 -0.10800  0.02652  0.16534  0.87225
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 1.094e+01  1.640e-01  66.697  < 2e-16 ***
## LENGTH.OF.STAY...ICU        4.668e-02  7.212e-03   6.472 1.05e-09 ***
## TOTAL.LENGTH.OF.STAY        3.616e-02  5.066e-03   7.138 2.86e-11 ***
## COST.OF.IMPLANT             8.568e-06  1.122e-06   7.633 1.74e-12 ***
## HR.PULSE                    1.367e-03  1.183e-03   1.155    0.250
## BODY.WEIGHT                 6.069e-03  1.109e-03   5.475 1.60e-07 ***
## MODE.OF.ARRIVALTRANSFERRED -1.757e-01  1.532e-01  -1.147    0.253
## MODE.OF.ARRIVALWALKED IN    1.278e-01  6.144e-02   2.080    0.039 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2437 on 165 degrees of freedom
## Multiple R-squared:  0.7899, Adjusted R-squared:  0.781
## F-statistic: 88.63 on 7 and 165 DF,  p-value: < 2.2e-16
```

```
vif(model15)
```

```
##                           GVIF Df GVIF^(1/(2*Df))
## LENGTH.OF.STAY...ICU 2.689199  1        1.639878
## TOTAL.LENGTH.OF.STAY 2.451576  1        1.565751
## COST.OF.IMPLANT      1.109584  1        1.053368
## HR.PULSE             1.668843  1        1.291837
## BODY.WEIGHT          1.951805  1        1.397070
## MODE.OF.ARRIVAL      1.280283  2        1.063718
```

```
model16 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
               TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+HR.PULSE+BODY.WEIGHT+AGE
             ,data = training)
summary(model16)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE + BODY.WEIGHT +
##     AGE, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.02686 -0.09310  0.02504  0.14753  0.91990
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          1.112e+01  1.487e-01  74.787  < 2e-16 ***
## LENGTH.OF.STAY...ICU 4.502e-02  7.245e-03   6.215 4.00e-09 ***
## TOTAL.LENGTH.OF.STAY 3.502e-02  5.257e-03   6.662 3.79e-10 ***
## COST.OF.IMPLANT      8.615e-06  1.145e-06   7.526 3.16e-12 ***
## HR.PULSE             1.149e-03  1.208e-03   0.951  0.34277
## BODY.WEIGHT          4.611e-03  1.749e-03   2.637  0.00916 **
## AGE                  7.461e-04  1.467e-03   0.508  0.61179
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2489 on 166 degrees of freedom
## Multiple R-squared:  0.7795, Adjusted R-squared:  0.7715
## F-statistic: 97.79 on 6 and 166 DF,  p-value: < 2.2e-16
```

```
vif(model16)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.600631             2.529615             1.106003
##             HR.PULSE          BODY.WEIGHT                  AGE
##             1.667464             4.654118             4.210183
```

```
model17 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
               TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+HR.PULSE+BODY.WEIGHT+BP..HIGH
             ,data = training)
summary(model17)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE + BODY.WEIGHT +
##     BP..HIGH, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.00733 -0.08260  0.02875  0.14480  0.88096
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)          1.094e+01  1.836e-01  59.590  < 2e-16 ***
## LENGTH.OF.STAY...ICU 4.488e-02  7.187e-03   6.245 3.43e-09 ***
## TOTAL.LENGTH.OF.STAY 3.602e-02  5.145e-03   7.000 6.04e-11 ***
## COST.OF.IMPLANT      8.757e-06  1.141e-06   7.676 1.33e-12 ***
## HR.PULSE             1.114e-03  1.197e-03   0.931 0.353405
## BODY.WEIGHT          4.362e-03  1.241e-03   3.515 0.000566 ***
## BP..HIGH             1.728e-03  1.113e-03   1.553 0.122276
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2473 on 166 degrees of freedom
## Multiple R-squared:  0.7823, Adjusted R-squared:  0.7744
## F-statistic: 99.41 on 6 and 166 DF,  p-value: < 2.2e-16
```

```
vif(model17)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.592718             2.454823             1.112729
##             HR.PULSE          BODY.WEIGHT             BP..HIGH
##             1.659346             2.374590             1.562895
```

```
model18 = lm(TOTAL.COST.TO.HOSPITAL~TOTAL.LENGTH.OF.STAY+
             COST.OF.IMPLANT+
             AGE+
             CREATININE+
             BP..HIGH+
             HR.PULSE+
             BODY.WEIGHT+
             BMI+
             IMPLANT.USED..Y.N.+
             TYPE.OF.ADMSN+
             KEY.COMPLAINTS..CODE+
             PAST.MEDICAL.HISTORY.CODE+
             MODE.OF.ARRIVAL,data = training)
summary(model18)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ TOTAL.LENGTH.OF.STAY +
##     COST.OF.IMPLANT + AGE + CREATININE + BP..HIGH + HR.PULSE +
##     BODY.WEIGHT + BMI + IMPLANT.USED..Y.N. + TYPE.OF.ADMSN +
##     KEY.COMPLAINTS..CODE + PAST.MEDICAL.HISTORY.CODE + MODE.OF.ARRIVAL,
##     data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.62138 -0.09968  0.00944  0.11689  0.81698
##
## Coefficients:
##                                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)                          1.056e+01  3.497e-01  30.203  < 2e-16
## TOTAL.LENGTH.OF.STAY                 5.301e-02  4.013e-03  13.211  < 2e-16
## COST.OF.IMPLANT                      6.083e-06  2.422e-06   2.511  0.01315
## AGE                                  1.223e-03  1.649e-03   0.741  0.45970
## CREATININE                           9.191e-02  6.621e-02   1.388  0.16725
## BP..HIGH                             1.592e-03  1.181e-03   1.349  0.17965
## HR.PULSE                             2.559e-03  1.294e-03   1.978  0.04990
## BODY.WEIGHT                          4.552e-03  2.783e-03   1.635  0.10419
## BMI                                 -3.487e-01  3.591e-01  -0.971  0.33325
## IMPLANT.USED..Y.N.                   2.470e-01  1.136e-01   2.175  0.03128
## TYPE.OF.ADMSN                        5.854e-02  2.675e-01   0.219  0.82712
## KEY.COMPLAINTS..CODEACHD            -9.396e-02  9.660e-02  -0.973  0.33237
## KEY.COMPLAINTS..CODECAD-DVD          9.504e-02  8.884e-02   1.070  0.28655
## KEY.COMPLAINTS..CODECAD-SVD         -1.960e-01  1.973e-01  -0.993  0.32226
## KEY.COMPLAINTS..CODECAD-TVD          1.679e-01  9.334e-02   1.798  0.07425
## KEY.COMPLAINTS..CODEOS-ASD           9.970e-02  9.622e-02   1.036  0.30194
## KEY.COMPLAINTS..CODEother- heart     4.811e-02  7.567e-02   0.636  0.52591
## KEY.COMPLAINTS..CODEother- respiratory  4.937e-02  9.240e-02   0.534  0.59397
## KEY.COMPLAINTS..CODEother-general   -9.010e-01  2.878e-01  -3.131  0.00212
## KEY.COMPLAINTS..CODEother-nervous    8.781e-04  1.848e-01   0.005  0.99622
## KEY.COMPLAINTS..CODEother-tertalogy  2.517e-01  1.148e-01   2.193  0.02991
## KEY.COMPLAINTS..CODEPM-VSD           2.801e-01  1.253e-01   2.235  0.02700
## KEY.COMPLAINTS..CODERHD             -6.448e-02  1.019e-01  -0.633  0.52804
## PAST.MEDICAL.HISTORY.CODEDiabetes1   1.443e-01  1.135e-01   1.272  0.20556
## PAST.MEDICAL.HISTORY.CODEDiabetes2   3.161e-01  1.228e-01   2.575  0.01107
## PAST.MEDICAL.HISTORY.CODEhypertension1  5.985e-02  8.221e-02   0.728  0.46782
## PAST.MEDICAL.HISTORY.CODEHypertension1  9.562e-02  1.838e-01   0.520  0.60381
## PAST.MEDICAL.HISTORY.CODEhypertension2  4.775e-02  9.055e-02   0.527  0.59880
## PAST.MEDICAL.HISTORY.CODEhypertension3  2.914e-02  1.348e-01   0.216  0.82908
## PAST.MEDICAL.HISTORY.CODEother      -9.128e-02  1.278e-01  -0.714  0.47612
## MODE.OF.ARRIVALTRANSFERRED          -2.479e-01  3.127e-01  -0.793  0.42937
## MODE.OF.ARRIVALWALKED IN             1.214e-01  2.673e-01   0.454  0.65037
##
## (Intercept)                         ***
## TOTAL.LENGTH.OF.STAY                ***
## COST.OF.IMPLANT                     *
## AGE
## CREATININE
## BP..HIGH
## HR.PULSE                            *
## BODY.WEIGHT
```

```
## BMI
## IMPLANT.USED..Y.N.                     *
## TYPE.OF.ADMSN
## KEY.COMPLAINTS..CODEACHD
## KEY.COMPLAINTS..CODECAD-DVD
## KEY.COMPLAINTS..CODECAD-SVD
## KEY.COMPLAINTS..CODECAD-TVD            .
## KEY.COMPLAINTS..CODEOS-ASD
## KEY.COMPLAINTS..CODEother- heart
## KEY.COMPLAINTS..CODEother- respiratory
## KEY.COMPLAINTS..CODEother-general      **
## KEY.COMPLAINTS..CODEother-nervous
## KEY.COMPLAINTS..CODEother-tertalogy    *
## KEY.COMPLAINTS..CODEPM-VSD             *
## KEY.COMPLAINTS..CODERHD
## PAST.MEDICAL.HISTORY.CODEDiabetes1
## PAST.MEDICAL.HISTORY.CODEDiabetes2     *
## PAST.MEDICAL.HISTORY.CODEhypertension1
## PAST.MEDICAL.HISTORY.CODEHypertension1
## PAST.MEDICAL.HISTORY.CODEhypertension2
## PAST.MEDICAL.HISTORY.CODEhypertension3
## PAST.MEDICAL.HISTORY.CODEother
## MODE.OF.ARRIVALTRANSFERRED
## MODE.OF.ARRIVALWALKED IN
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2388 on 141 degrees of freedom
## Multiple R-squared:  0.8276, Adjusted R-squared:  0.7897
## F-statistic: 21.84 on 31 and 141 DF,  p-value: < 2.2e-16
```

```
vif(model17)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.592718             2.454823             1.112729
##             HR.PULSE          BODY.WEIGHT             BP..HIGH
##             1.659346             2.374590             1.562895
```

```
model18 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
               TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+HR.PULSE+BODY.WEIGHT+CREATININE
             ,data = training)
summary(model18)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE + BODY.WEIGHT +
##     CREATININE, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.03177 -0.09028  0.03292  0.14416  0.92144
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)          1.111e+01  1.485e-01  74.825  < 2e-16 ***
## LENGTH.OF.STAY...ICU 4.508e-02  7.242e-03   6.225 3.81e-09 ***
## TOTAL.LENGTH.OF.STAY 3.522e-02  5.208e-03   6.762 2.22e-10 ***
## COST.OF.IMPLANT      8.583e-06  1.142e-06   7.513 3.40e-12 ***
## HR.PULSE             1.115e-03  1.205e-03   0.925   0.3564
## BODY.WEIGHT          5.017e-03  1.259e-03   3.986   0.0001 ***
## CREATININE           2.472e-02  5.394e-02   0.458   0.6474
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.249 on 166 degrees of freedom
## Multiple R-squared:  0.7794, Adjusted R-squared:  0.7714
## F-statistic: 97.75 on 6 and 166 DF,  p-value: < 2.2e-16
```

```
vif(model18)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.597842             2.482518             1.101320
##             HR.PULSE          BODY.WEIGHT           CREATININE
##             1.659688             2.411205             1.724698
```

```
model19 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
               TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+HR.PULSE+BODY.WEIGHT+KEY.COMPLAINTS..CODE
             ,data = training)
summary(model19)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE + BODY.WEIGHT +
##     KEY.COMPLAINTS..CODE, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.66444 -0.10827  0.01482  0.13556  0.82660
##
## Coefficients:
##                                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        1.109e+01  1.497e-01  74.083  < 2e-16
## LENGTH.OF.STAY...ICU               4.304e-02  6.952e-03   6.191 5.13e-09
## TOTAL.LENGTH.OF.STAY               3.221e-02  4.976e-03   6.472 1.20e-09
## COST.OF.IMPLANT                    8.037e-06  1.377e-06   5.835 3.04e-08
## HR.PULSE                           1.351e-03  1.139e-03   1.186  0.23731
## BODY.WEIGHT                        4.422e-03  1.380e-03   3.204  0.00164
## KEY.COMPLAINTS..CODEACHD          -1.382e-01  8.681e-02  -1.592  0.11333
## KEY.COMPLAINTS..CODECAD-DVD        1.704e-01  7.065e-02   2.412  0.01703
## KEY.COMPLAINTS..CODECAD-SVD        1.134e-01  1.716e-01   0.661  0.50976
## KEY.COMPLAINTS..CODECAD-TVD        2.360e-01  7.284e-02   3.240  0.00146
## KEY.COMPLAINTS..CODEOS-ASD         4.996e-02  8.536e-02   0.585  0.55919
## KEY.COMPLAINTS..CODEother- heart   4.868e-02  6.690e-02   0.728  0.46789
## KEY.COMPLAINTS..CODEother- respiratory  4.006e-02  8.428e-02   0.475  0.63520
## KEY.COMPLAINTS..CODEother-general -9.269e-01  2.298e-01  -4.034 8.60e-05
## KEY.COMPLAINTS..CODEother-nervous -4.017e-02  1.707e-01  -0.235  0.81424
## KEY.COMPLAINTS..CODEother-tertalogy  2.622e-01  1.042e-01   2.516  0.01290
## KEY.COMPLAINTS..CODEPM-VSD         2.204e-01  1.122e-01   1.964  0.05128
## KEY.COMPLAINTS..CODERHD            1.377e-01  8.724e-02   1.579  0.11648
##
## (Intercept)                        ***
## LENGTH.OF.STAY...ICU               ***
## TOTAL.LENGTH.OF.STAY               ***
## COST.OF.IMPLANT                    ***
## HR.PULSE
## BODY.WEIGHT                        **
## KEY.COMPLAINTS..CODEACHD
## KEY.COMPLAINTS..CODECAD-DVD        *
## KEY.COMPLAINTS..CODECAD-SVD
## KEY.COMPLAINTS..CODECAD-TVD        **
## KEY.COMPLAINTS..CODEOS-ASD
## KEY.COMPLAINTS..CODEother- heart
## KEY.COMPLAINTS..CODEother- respiratory
## KEY.COMPLAINTS..CODEother-general  ***
## KEY.COMPLAINTS..CODEother-nervous
## KEY.COMPLAINTS..CODEother-tertalogy  *
## KEY.COMPLAINTS..CODEPM-VSD         .
## KEY.COMPLAINTS..CODERHD
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.223 on 155 degrees of freedom
## Multiple R-squared:  0.8347, Adjusted R-squared:  0.8166
## F-statistic: 46.06 on 17 and 155 DF,  p-value: < 2.2e-16
```

```
vif(model19)
```

```
##                           GVIF Df GVIF^(1/(2*Df))
## LENGTH.OF.STAY...ICU 2.984175  1        1.727476
## TOTAL.LENGTH.OF.STAY 2.824702  1        1.680685
## COST.OF.IMPLANT      1.995340  1        1.412565
## HR.PULSE             1.847427  1        1.359201
## BODY.WEIGHT          3.612569  1        1.900676
## KEY.COMPLAINTS..CODE 7.480789 12        1.087463
```

```
model20 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
               TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+HR.PULSE+BODY.WEIGHT+KEY.COMPLAINTS..CODE
+
               PAST.MEDICAL.HISTORY.CODE + IMPLANT.USED..Y.N.+AGE,data = training)
summary(model20)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE + BODY.WEIGHT +
##     KEY.COMPLAINTS..CODE + PAST.MEDICAL.HISTORY.CODE + IMPLANT.USED..Y.N. +
##     AGE, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.66256 -0.08499  0.00451  0.12000  0.67651
##
## Coefficients:
##                                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)                          1.109e+01  1.526e-01  72.683  < 2e-16
## LENGTH.OF.STAY...ICU                 3.831e-02  7.020e-03   5.458 2.02e-07
## TOTAL.LENGTH.OF.STAY                 3.248e-02  5.036e-03   6.450 1.54e-09
## COST.OF.IMPLANT                      4.928e-06  2.195e-06   2.245 0.026272
## HR.PULSE                             1.539e-03  1.180e-03   1.304 0.194187
## BODY.WEIGHT                          3.115e-03  1.878e-03   1.659 0.099349
## KEY.COMPLAINTS..CODEACHD            -1.430e-01  8.686e-02  -1.646 0.101856
## KEY.COMPLAINTS..CODECAD-DVD          1.250e-01  8.062e-02   1.550 0.123237
## KEY.COMPLAINTS..CODECAD-SVD          1.977e-02  1.814e-01   0.109 0.913366
## KEY.COMPLAINTS..CODECAD-TVD          1.620e-01  8.250e-02   1.964 0.051416
## KEY.COMPLAINTS..CODEOS-ASD           6.176e-02  8.508e-02   0.726 0.469058
## KEY.COMPLAINTS..CODEother- heart     4.360e-02  6.712e-02   0.650 0.516979
## KEY.COMPLAINTS..CODEother- respiratory  4.262e-02  8.424e-02   0.506 0.613656
## KEY.COMPLAINTS..CODEother-general   -8.813e-01  2.597e-01  -3.393 0.000889
## KEY.COMPLAINTS..CODEother-nervous   -3.510e-02  1.692e-01  -0.207 0.835994
## KEY.COMPLAINTS..CODEother-tertalogy  2.702e-01  1.046e-01   2.584 0.010754
## KEY.COMPLAINTS..CODEPM-VSD           2.172e-01  1.119e-01   1.941 0.054181
## KEY.COMPLAINTS..CODERHD              6.522e-02  9.426e-02   0.692 0.490072
## PAST.MEDICAL.HISTORY.CODEDiabetes1   8.195e-02  1.041e-01   0.787 0.432449
## PAST.MEDICAL.HISTORY.CODEDiabetes2   2.703e-01  1.064e-01   2.539 0.012148
## PAST.MEDICAL.HISTORY.CODEhypertension1  5.576e-02  7.431e-02   0.750 0.454199
## PAST.MEDICAL.HISTORY.CODEHypertension1  1.083e-01  1.680e-01   0.645 0.520059
## PAST.MEDICAL.HISTORY.CODEhypertension2  1.612e-02  7.859e-02   0.205 0.837789
## PAST.MEDICAL.HISTORY.CODEhypertension3  4.894e-02  1.239e-01   0.395 0.693342
## PAST.MEDICAL.HISTORY.CODEother      -5.364e-02  1.180e-01  -0.455 0.650047
## IMPLANT.USED..Y.N.                   2.073e-01  1.048e-01   1.979 0.049727
## AGE                                  1.111e-03  1.438e-03   0.773 0.440947
##
## (Intercept)                         ***
## LENGTH.OF.STAY...ICU                ***
## TOTAL.LENGTH.OF.STAY                ***
## COST.OF.IMPLANT                     *
## HR.PULSE
## BODY.WEIGHT                         .
## KEY.COMPLAINTS..CODEACHD
## KEY.COMPLAINTS..CODECAD-DVD
## KEY.COMPLAINTS..CODECAD-SVD
## KEY.COMPLAINTS..CODECAD-TVD         .
## KEY.COMPLAINTS..CODEOS-ASD
## KEY.COMPLAINTS..CODEother- heart
## KEY.COMPLAINTS..CODEother- respiratory
## KEY.COMPLAINTS..CODEother-general   ***
```

```
## KEY.COMPLAINTS..CODEother-nervous
## KEY.COMPLAINTS..CODEother-tertalogy    *
## KEY.COMPLAINTS..CODEPM-VSD             .
## KEY.COMPLAINTS..CODERHD
## PAST.MEDICAL.HISTORY.CODEDiabetes1
## PAST.MEDICAL.HISTORY.CODEDiabetes2     *
## PAST.MEDICAL.HISTORY.CODEhypertension1
## PAST.MEDICAL.HISTORY.CODEHypertension1
## PAST.MEDICAL.HISTORY.CODEhypertension2
## PAST.MEDICAL.HISTORY.CODEhypertension3
## PAST.MEDICAL.HISTORY.CODEother
## IMPLANT.USED..Y.N.                     *
## AGE
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2206 on 146 degrees of freedom
## Multiple R-squared:  0.8477, Adjusted R-squared:  0.8205
## F-statistic: 31.24 on 26 and 146 DF,  p-value: < 2.2e-16
```

```
vif(model20)
```

```
##                               GVIF Df GVIF^(1/(2*Df))
## LENGTH.OF.STAY...ICU       3.108701  1        1.763151
## TOTAL.LENGTH.OF.STAY       2.955731  1        1.719224
## COST.OF.IMPLANT            5.177365  1        2.275382
## HR.PULSE                   2.026393  1        1.423514
## BODY.WEIGHT                6.837625  1        2.614885
## KEY.COMPLAINTS..CODE      27.191201 12        1.147540
## PAST.MEDICAL.HISTORY.CODE  6.328080  7        1.140864
## IMPLANT.USED..Y.N.         6.163160  1        2.482571
## AGE                        5.144917  1        2.268241
```

```
model21 = lm(TOTAL.COST.TO.HOSPITAL~LENGTH.OF.STAY...ICU+
               TOTAL.LENGTH.OF.STAY+COST.OF.IMPLANT+HR.PULSE+BODY.WEIGHT+AGE
             ,data = training)
summary(model21)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE + BODY.WEIGHT +
##     AGE, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.02686 -0.09310  0.02504  0.14753  0.91990
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)          1.112e+01  1.487e-01  74.787  < 2e-16 ***
## LENGTH.OF.STAY...ICU 4.502e-02  7.245e-03   6.215 4.00e-09 ***
## TOTAL.LENGTH.OF.STAY 3.502e-02  5.257e-03   6.662 3.79e-10 ***
## COST.OF.IMPLANT      8.615e-06  1.145e-06   7.526 3.16e-12 ***
## HR.PULSE             1.149e-03  1.208e-03   0.951  0.34277
## BODY.WEIGHT          4.611e-03  1.749e-03   2.637  0.00916 **
## AGE                  7.461e-04  1.467e-03   0.508  0.61179
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2489 on 166 degrees of freedom
## Multiple R-squared:  0.7795, Adjusted R-squared:  0.7715
## F-statistic: 97.79 on 6 and 166 DF,  p-value: < 2.2e-16
```

```
vif(model21)
```

```
## LENGTH.OF.STAY...ICU TOTAL.LENGTH.OF.STAY      COST.OF.IMPLANT
##             2.600631             2.529615             1.106003
##             HR.PULSE          BODY.WEIGHT                  AGE
##             1.667464             4.654118             4.210183
```

```
model22 = lm(TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +AGE+
                   TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE +
                   BODY.WEIGHT + PAST.MEDICAL.HISTORY.CODE,
                 data = training)
summary(model22)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ LENGTH.OF.STAY...ICU +
##     AGE + TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE +
##     BODY.WEIGHT + PAST.MEDICAL.HISTORY.CODE, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.83989 -0.08646  0.03387  0.13132  0.69043
##
## Coefficients:
##                                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)                          1.117e+01  1.473e-01  75.870  < 2e-16
## LENGTH.OF.STAY...ICU                 3.801e-02  7.177e-03   5.296 3.89e-07
## AGE                                  7.358e-04  1.505e-03   0.489 0.625607
## TOTAL.LENGTH.OF.STAY                 3.642e-02  5.096e-03   7.147 3.03e-11
## COST.OF.IMPLANT                      9.257e-06  1.159e-06   7.988 2.60e-13
## HR.PULSE                             1.076e-03  1.204e-03   0.893 0.372994
## BODY.WEIGHT                          2.759e-03  1.784e-03   1.547 0.123887
## PAST.MEDICAL.HISTORY.CODEDiabetes1   1.741e-01  1.025e-01   1.699 0.091227
## PAST.MEDICAL.HISTORY.CODEDiabetes2   3.808e-01  1.047e-01   3.637 0.000372
## PAST.MEDICAL.HISTORY.CODEhypertension1  1.039e-01  7.171e-02   1.449 0.149230
## PAST.MEDICAL.HISTORY.CODEHypertension1  1.372e-01  1.788e-01   0.767 0.444210
## PAST.MEDICAL.HISTORY.CODEhypertension2  5.253e-02  8.242e-02   0.637 0.524842
## PAST.MEDICAL.HISTORY.CODEhypertension3  2.256e-02  1.281e-01   0.176 0.860419
## PAST.MEDICAL.HISTORY.CODEother         -2.431e-01  1.124e-01  -2.164 0.031984
##
## (Intercept)                          ***
## LENGTH.OF.STAY...ICU                 ***
## AGE
## TOTAL.LENGTH.OF.STAY                 ***
## COST.OF.IMPLANT                      ***
## HR.PULSE
## BODY.WEIGHT
## PAST.MEDICAL.HISTORY.CODEDiabetes1   .
## PAST.MEDICAL.HISTORY.CODEDiabetes2   ***
## PAST.MEDICAL.HISTORY.CODEhypertension1
## PAST.MEDICAL.HISTORY.CODEHypertension1
## PAST.MEDICAL.HISTORY.CODEhypertension2
## PAST.MEDICAL.HISTORY.CODEhypertension3
## PAST.MEDICAL.HISTORY.CODEother       *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2396 on 159 degrees of freedom
## Multiple R-squared:  0.8043, Adjusted R-squared:  0.7883
## F-statistic: 50.26 on 13 and 159 DF,  p-value: < 2.2e-16
```

```
vif(model22)
```

```
##                                GVIF Df GVIF^(1/(2*Df))
## LENGTH.OF.STAY...ICU      2.754394  1        1.659637
## AGE                       4.780166  1        2.186359
## TOTAL.LENGTH.OF.STAY      2.565460  1        1.601705
## COST.OF.IMPLANT           1.223208  1        1.105987
## HR.PULSE                  1.787308  1        1.336902
## BODY.WEIGHT               5.226027  1        2.286051
## PAST.MEDICAL.HISTORY.CODE 2.338618  7        1.062562
```

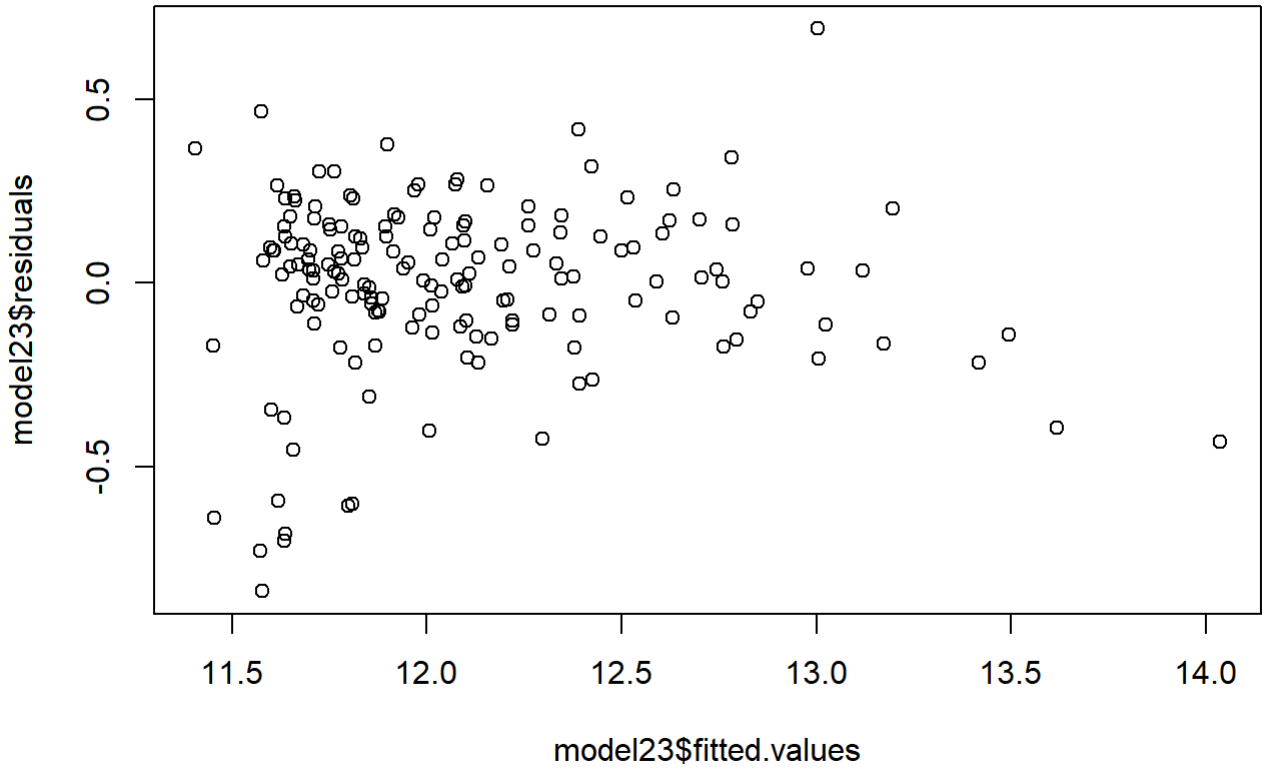
```
model23 = lm(TOTAL.COST.TO.HOSPITAL ~ AGE+LENGTH.OF.STAY..WARD+
               TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE +
               BODY.WEIGHT + PAST.MEDICAL.HISTORY.CODE,
             data = training)
summary(model23)
```

```
##
## Call:
## lm(formula = TOTAL.COST.TO.HOSPITAL ~ AGE + LENGTH.OF.STAY..WARD +
##     TOTAL.LENGTH.OF.STAY + COST.OF.IMPLANT + HR.PULSE + BODY.WEIGHT +
##     PAST.MEDICAL.HISTORY.CODE, data = training)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.84010 -0.08865  0.03293  0.13618  0.69296
##
## Coefficients:
##                                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)                          1.117e+01  1.472e-01  75.858  < 2e-16
## AGE                                  6.018e-04  1.509e-03   0.399 0.690625
## LENGTH.OF.STAY..WARD                -3.718e-02  7.082e-03  -5.250 4.81e-07
## TOTAL.LENGTH.OF.STAY                 7.422e-02  4.964e-03  14.951  < 2e-16
## COST.OF.IMPLANT                      9.262e-06  1.160e-06   7.983 2.69e-13
## HR.PULSE                             1.096e-03  1.205e-03   0.909 0.364595
## BODY.WEIGHT                          2.893e-03  1.785e-03   1.621 0.107035
## PAST.MEDICAL.HISTORY.CODEDiabetes1   1.743e-01  1.026e-01   1.699 0.091320
## PAST.MEDICAL.HISTORY.CODEDiabetes2   3.871e-01  1.046e-01   3.700 0.000297
## PAST.MEDICAL.HISTORY.CODEhypertension1  1.042e-01  7.180e-02   1.451 0.148812
## PAST.MEDICAL.HISTORY.CODEHypertension1  1.405e-01  1.791e-01   0.784 0.433915
## PAST.MEDICAL.HISTORY.CODEhypertension2  4.525e-02  8.251e-02   0.548 0.584233
## PAST.MEDICAL.HISTORY.CODEhypertension3  2.297e-02  1.283e-01   0.179 0.858085
## PAST.MEDICAL.HISTORY.CODEother      -2.445e-01  1.125e-01  -2.173 0.031245
##
## (Intercept)                         ***
## AGE
## LENGTH.OF.STAY..WARD                ***
## TOTAL.LENGTH.OF.STAY                ***
## COST.OF.IMPLANT                     ***
## HR.PULSE
## BODY.WEIGHT
## PAST.MEDICAL.HISTORY.CODEDiabetes1  .
## PAST.MEDICAL.HISTORY.CODEDiabetes2  ***
## PAST.MEDICAL.HISTORY.CODEhypertension1
## PAST.MEDICAL.HISTORY.CODEHypertension1
## PAST.MEDICAL.HISTORY.CODEhypertension2
## PAST.MEDICAL.HISTORY.CODEhypertension3
## PAST.MEDICAL.HISTORY.CODEother      *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2399 on 159 degrees of freedom
## Multiple R-squared:  0.8038, Adjusted R-squared:  0.7877
## F-statistic:  50.1 on 13 and 159 DF,  p-value: < 2.2e-16
```

```
vif(model23)
```

```
##                              GVIF Df GVIF^(1/(2*Df))
## AGE                      4.794039  1        2.189529
## LENGTH.OF.STAY..WARD     2.125226  1        1.457815
## TOTAL.LENGTH.OF.STAY     2.428377  1        1.558325
## COST.OF.IMPLANT          1.223148  1        1.105960
## HR.PULSE                 1.786144  1        1.336467
## BODY.WEIGHT              5.219414  1        2.284604
## PAST.MEDICAL.HISTORY.CODE 2.324487  7        1.062102
```

```
plot(model23$fitted.values,model23$residuals)
```




```
#test the model on testing data
prediction = predict(model23,testing)
head(prediction)
```

```
##        6        9       13       14       15       22
## 12.96734 13.20720 14.33053 12.63198 12.63808 12.50184
```

```
testing$prediction = prediction
head(testing$TOTAL.COST.TO.HOSPITAL)
```

```
## [1] 12.82761 12.98890 13.47529 12.94811 13.11427 12.57404
```

```
plot(testing$TOTAL.COST.TO.HOSPITAL,type="l",col="green")
lines(prediction,type="l",col="blue")
```

